

ISSN 2166-6768 (print)
ISSN 2166-6792 (online)



Journal of Enterprise Architecture

2016, Volume 12, Number 4

From the Editor
Leonard Fehskens

Short Subject: Don't Use the A Word!
Alberto S. Llanes, PhD and Maynard Avery Austin III, Esq.

Different Approaches to Enterprise Architecture
Svyatoslav Kotusev

Theoretical Perspectives of Enterprise Architecture for Technological Transformation
Torben Tambo

From Enterprise Architect to Opportunity Architect: The Changing Role of Enterprise Architecture in a Digital Transformation Context
Greet Bontinck, Prof. Dr. Bjorn Cumps, Prof. Dr. Stijn Viaene, Dr. Wesley Bille, and Joachim Vanden Brande

Short Subject: Architectural Roles in an "Agile Landscape"
Phil Bowker, Jacques Colle, Kenneth van Rumste

Book Review:
Software Design Decoded: 66 Ways Experts Think, by Marian Petre & André van der Hoek, with Illustrations by Yen Quach, and
Seventy-nine Short Essays on Design, by Michael Bierut
Reviewed by Leonard Fehskens

A Conceptual Framework for a Professional Code of Ethics for Enterprise Architects
Leonard Fehskens

Journal of Enterprise Architecture®

CEO, Association of Enterprise Architects: Chris Forde

Chief Editor: Leonard Fehskens, Association of Enterprise Architects

Production Editor: Cathy Fox, The Open Group

Associate Editors

Tyson Brooks, PhD, PMP Co-Director, Syracuse University	William W. Krauter, PhD Senior Architect, Lockheed Martin Corporation
Dick Burk Enterprise Architect	Haiping Luo, PhD International Trade Administration, US Dept. of Commerce
Brian Cameron, PhD Associate Dean & Clinical Professor, PA State University	Stephen Marley, PhD Jeffries Technology Solutions, Inc.
Gary Doucet Human Resources and Skills Development Canada	George Paras Managing Director, EADirections
Robert Ellinger, PhD Enterprise Architect	Pallab Saha, PhD Professor of Information Systems, National University of Singapore
Steve Else, PhD CEO, EA Principals & Adjunct Professor, University of Denver	P. Kathie Sowell President, Custom Enterprise Solutions, LLC/SowellEAC
William A. Estrem, PhD President, Metaplexity Associates, Inc.	Eskil Swende Partner IRM Sweden, President of DAMA Scandinavia
Jups Heikkilä, PhD Professor, University of Turku	Torben Tambo Associate Professor, Aarhus University Institute of Business & Technology
Leon Kappelman, PhD College of Business, University of North Texas	Tim Westbrock Managing Director, EADirections

About the Journal: The Journal of Enterprise Architecture (JEA) is published by the Association of Enterprise Architects, 548 Market St. #54820, San Francisco, CA 94104-5401, USA, Tel: +1 781 564-9200, www.globalaea.org. The JEA is a peer-reviewed international publication for the EA community. Four issues are published each year. The JEA supports the global practitioner community of interest through the publication of articles that promote the profession of enterprise architecture, and deals with issues regarding practices and methods, case studies, and standards at the national and international levels. The views expressed in JEA articles are those of the respective authors, and not necessarily those of the publisher, the Chief Editor, the associate editors, or the Association of Enterprise Architects (AEA).

Copyright: © 2005-2016 Association of Enterprise Architects. The reproduction, storage, or transmission of JEA articles or other content is prohibited without prior permission in writing from the JEA Chief Editor, who can be contacted via email at l.fehskens@globalaea.org.

Trademarks: Association of Enterprise Architects®, the AEA logo, and Journal of Enterprise Architecture® are registered trademarks of the Association of Enterprise Architects.

Article Submissions: Authors may submit properly formatted manuscripts to the JEA Chief Editor at journal.submissions@globalaea.org for publication consideration. Author submission guidelines are available on the AEA website at www.globalaea.org. Copyright of all accepted articles and other published content is transferred by the author to the JEA upon notification of acceptance by the JEA Chief Editor.

Subscription: Available to AEA members. The annual cost of AEA membership is US\$75.00. To join, please refer to www.globalaea.org. Online edition: no charge access for AEA members.

Back Issues: All back issues of the JEA are available in electronic form, and can be accessed online by AEA members. Libraries can apply for access by contacting the AEA.

Contents

From the Editor, Leonard Fehskens	4
Short Subject: Don't Use the A Word!.....	6
Article: Different Approaches to Enterprise Architecture	9
Article: Theoretical Perspectives of Enterprise Architecture for Technological Transformation	18
Article: From Enterprise Architect to Opportunity Architect: The Changing Role of Enterprise Architecture in a Digital Transformation Context.....	32
Short Subject: Architectural Roles in an “Agile Landscape”	42
Book Review:	
Software Design Decoded: 66 Ways Experts Think, by Marian Petre & André van der Hoek, with Illustrations by Yen Quach and Seventy-nine Short Essays on Design, by Michael Bierut	47
Article: Conceptual Framework for a Professional Code of Ethics for Enterprise Architects.....	49

From the Editor

Leonard Fehskens

Welcome to the December 2016 issue of the Journal of Enterprise Architecture.

The issue opens with a discussion of a well-intentioned admonition that we often hear – “Don’t Use the A Word”. As Jeff Scott so cogently observes, our clients’ eyes may glaze over at the mention of the word “architecture”, but they are generally enthusiastic consumers of the ultimate results of architectural thinking. But, as Alberto Llanes and Maynard Austin argue in their Short Subject, it is architecture that makes the delivery of that value possible, and it is what defines us as a profession.

The three articles that follow explore that foundation in more depth. Svyatoslav Kotusev’s “Different Approaches to Enterprise Architecture” takes an historically informed perspective on the various ways we have come to think about enterprise architecture. Torben Tambo’s “Theoretical Perspectives of Enterprise Architecture for Technological Transformation” offers an insightful connection between enterprise architecture and the discipline of Management of Technology (MOT). Shifting the focus from theory to practice, Greet Bontinck et al’s “From Enterprise Architect to Opportunity Architect: The Changing Role of Enterprise Architecture in a Digital Transformation Context” looks at how enterprise architecture actually contributes to the “digital transformation” of contemporary enterprises via multiple case studies.

Such digital transformations are often at least implicitly driven by the need for agility, and the Short Subject “Architectural Roles in an ‘Agile Landscape’” by Phil Bowker et al. looks at how architects can provide architectural coherence in an agile development environment.

In this issue’s Book Review I consider two books that sound alike but turn out to be quite different in detail. One is about software design, and the other is about graphic design, but they can each provide value, in different ways, to the reflective practitioner of enterprise architecture.

Finally, in support of the AEA’s initiative to develop a professional code of ethics for practicing enterprise architects, I survey the literature on professional codes of ethics in a way that I trust calls attention to the essential issues we as a profession must address in such a code.

As always, I urge you to consider contributing to the Journal. You don’t have to write something to do so – you can suggest something you’ve seen published elsewhere that you think would be of value to the enterprise architecture community if it were featured in the JEA. If you do want to write something original, we’re now accepting non-peer-reviewed short subjects (a few pages) and articles (a half dozen or so pages), as well as the traditional peer-reviewed longer articles. We’re especially interested in business and other forms of enterprise-related material, as well as technological topics. Finally, you can always send us your own “Four Questions”, suggest someone you’d like to have submit their “Four Questions”, volunteer to “Talk Shop” with me, or suggest someone you’d like me to “Talk Shop” with.

Len.

Visit the AEA website at www.globalaea.org



The screenshot shows the AEA website homepage. At the top left is the AEA logo with the text "Association of Enterprise Architects®". To the right is a "Translate this page" widget for Microsoft Translator, set to Spanish. Below the header is a navigation bar with links: "CONTACT US", "SIGN IN", and "JOIN". On the left is a green sidebar menu with the following items: "Home", "About Us", "AEA Open Badges", "On Demand Events", "Journal of EA", "Community Blogs", "Get Involved", "Groups", "Find a Group", "Forums", "Career Center", "AEA Membership", "Join a Chapter", and "Who's Online Now?". The main content area features a large photo of a conference room. Below the photo is the heading "Welcome to our Association of Enterprise Architects Online Community!" followed by a welcome message and a sign-in section. The sign-in section includes buttons for "Login with Facebook" and "Login with LinkedIn", a "SIGN IN" button, and links for "Forgot your password?" and "Haven't joined yet?". On the right side of the main content area is a "Latest News" section with three items: "11/18/2016 Good e-Learning 10% Discount for AEA Members", "10/6/2016 Journal of Enterprise Architecture 2016, Issue No. 3 now available!", and "9/13/2016 Chris Forde appointed CEO of the AEA".

Call for Papers

The Journal of Enterprise Architecture is accepting article submissions for its future issues. Research and best practice articles are sought on enterprise architecture-related topics, including:

- Case Studies, Configuration Management, Culture, Documentation
- Evaluation, Frameworks, Governance, Implementation, Maintenance
- Methodologies, Taxonomies, Theory, Training, Tools, Use, Value

Please send articles to the JEA Chief Editor at journal.submissions@globalaea.org. Author submission guidelines can be found on the AEA website at www.globalaea.org.

Short Subject

Don't Use the A Word!

Alberto S. Llanes, PhD and Maynard Avery Austin III, Esq.

Abstract

Enterprise architecture's diversity is both its strength and weakness. We must unify as a community for both the sake of our chosen profession and the benefit of our stakeholders.

Keywords

Architecture, enterprise architecture, diversity, culture, unity

INTRODUCTION

It's not unusual to hear someone advise, "Don't use the A word!". The first time we heard this we thought, to what could they be referring? Imagine our shock when we learned this was being said by professional peers in reference to our own discipline – architecture! However, rather than blame the naysayers of our craft for their ignorance, let's look within, as perhaps this lack of credibility was of our own making.

A CULTURAL DIVIDE: DIVERSITY

Unlike the discipline of project management, where the Project Management Institute is one organization, with one body of knowledge, one fundamental certification, and one dominant tool, the discipline of enterprise architecture is diverse. We have no central certification authority so our certifications are confusing to anyone outside our discipline, and this makes it possible for people to claim professional status they don't deserve. There are varied approaches: integrated, layered, or segmented. There are different areas of specialization: strategic, business, systems, data, and technological infrastructure. There are frameworks and tools galore! There are dogmatic purists and people with less narrowly bounded ideas of the scope of the discipline. There are practitioners who are solutions-driven and those who would simply model the world for its own sake. One problem with some architects seeing integrated architectures, others segments, and others layers, is that they often only see the view that they conceptually live in. Notation methods are as diverse as the open source developer community, whereas others practice no notation at all. Many systems architects perceive business architects as creating shelfware, whereas many business architects perceive systems architects as commodities. The only commonality seems to be that, for each of us, our way is the only one right way, and only we are true enterprise architects!

A CULTURAL DIVIDE: STRENGTHS

The palette from which we can choose how to investigate, model, communicate, and guide customers at multiple layers of abstraction or depth is diverse. We can literally cherry pick the best tools for the job. For example, the Zachman Framework with its interrogatives and approach for gaining understanding and communicating at the appropriate level provides a powerful taxonomy. The Open Group TOGAF® Architecture Development Model (ADM) is a proven approach to architectural success. The Federal Enterprise Architecture Framework (FEAF) excels at scalable resource planning and its segment approach to gaining cohesive understanding across layers for problem solving. The Department of Defense Architectural Framework (DoDAF) excels in integrated views of enterprise operations that link operations from the finest detail to the highest-level abstraction.

From within our discipline the tools exist to not only advise our clients but also design a brave new world, whether from strategic planning through business modeling, use-case development and UML® application development through information modeling. But architecture for architecture's sake has no purpose. Whether to learn, advise, plan, or solve problems, it must have tangible value. Tangible architecture? There's the rub. It's not a tangible domain. Perceptions of value often elude people when contributing factors are once or twice removed. People see the services, the project deliverables, the bits and the bytes, but not the conceptual planning that conceived and guided it. Although everyone sees the fire, few can trace it all the way back to the spark that catalyzed the system. But wait, didn't we state our palette, our diversity, is our strength? Yes, we did. Our problem isn't technical, it's cultural.

ACHIEVING UNITY: HOLISTIC PERSPECTIVES

Our profession in large part suffers from near-sightedness. Systems architects, business architects, and data architects all see value in what's directly in front of them, but not in what other architects are doing elsewhere in the enterprise. This incomplete perspective taints our profession from within. It is essential as architects that we consider the beginning, end, and intervening connections that integrate the enterprises' uppermost leaves to its roots, seeing the complete tree and then its place in the forest. Otherwise we can only enable organic growth and, while it and mutation occasionally yield new fruit, they just as easily lead to the deterioration of the organism. Our enterprises can ill afford lack of planning or incomplete architectural vision.

Only by envisioning and at least conceptually integrating the big picture can we then communicate it effectively in whatever level of detail or abstraction is needed for the view of a particular stakeholder. Doing less too often communicates misleading if not incorrect abstractions to stakeholders, undermining their contributions and the integrity of our profession.

ACHIEVING UNITY: GUARDING THE GATE

Trying to make sense of a confusing job description, people are often hired based on which "tools" they've used before. This makes about as much sense as one of the authors of this paper claiming they were the new Lord Byron or John Maynard Keynes because they know their way around a word processor or spreadsheet application.

Misuse of the "architect" title leads to well-intentioned recruiters thinking of the role as "technical", and what should be actual architecture positions are often filled by technical people who lack architectural skills, knowledge, and experience. And while they do the best that they can, they inevitably eventually clash with actual members of the profession. However, the blame cannot be put on human resource management as, without a single authority like the Project Management Institute, too many certifications makes it difficult to tell which ones matter. Nonetheless, it is good practice for an architect to hold at least one of the many existing enterprise architecture certifications as then we would have a reasonable expectation of their having a clue regarding the discipline and not being another generic "technical person".

ACHIEVING UNITY: BRIDGING NOT ALIENATING

There is an endearing but unfortunately accurate joke regarding architects, "What is the difference between a terrorist and an enterprise architect? ... You can negotiate with a terrorist". This characterization, and its

basis in common behaviors, is unfortunately anathema to those who stand the most to reap the value of our profession. If you cannot see the big picture, if only your view is the "correct" view, if you cannot see the value in others' work, then please retire or learn to listen and compromise. Business is a domain of clear communications, negotiations, plans, and execution – it does not benefit from conflict. As enterprise architects we bridge layers and cultures, building segments, abstracting and decomposing views so they communicate accurately and at the proper level for the stakeholder who most stands to benefit from the information. If the above jest describes you, and you must be doggedly adversarial, please find the nearest dojo or boxing ring and work it out there. Your professional peers will be most appreciative.

UNITY: YOU HAVE NO EXCUSES MOVING FORWARD

If you've read this far you are now well informed on the situation facing our honorable profession and the ill fate of our stakeholders should we fail in our mission. Say the "A" word, live it, breathe it, love it, work it, and let us proceed to **Architect** the enterprise into increased safety, security, efficiency, and productivity. If we fail in our mission those three fingers will be pointing back at you, the enterprise **Architect**. Failure is not an option.

ABOUT THE AUTHORS

Alberto S. Llanes PhD is a health information technology professional and CEA whose experience ranges from hardware to all facets of software design, development and management, systems and enterprise architecture. Dr. Llanes holds TOGAF® 8 & 9, FEAC CEA, and PMP Certifications. Dr. Llanes may be reached at AlbertoSalustianoLlanes@gmail.com.

Maynard Avery Austin III, Esq. is a FEAC CEA whose background includes serving as an Enterprise Architect supporting the Commander Naval Installations Command (CNIC) Chief Information Officer (CIO) and the Secretary of the Air Force (SAF) CIO. Mr. Austin may be reached at maveryaustin@gmail.com.

Good e-Learning

15%
Discount For All
AEA Members

At checkout enter Code: AEA15

Good e-Learning offer a suite of interactive
e-learning solutions for Enterprise Architecture.
Completely online, you can study when it suits
you most – at home, at work or on the move.

Courses include:

TOGAF® Certified

IT4IT™ Foundation

Awareness: TOGAF®

Awareness: IT4IT™

Implementation: TOGAF®

ArchiMate® Certified

www.goodelearning.com

Offer ends, April 30th 2017

TOGAF® 9 Foundation (level 1) v9.1, TOGAF® 9 Certified (level 2) v9.1, TOGAF® 9 Certification (level 1 & 2) v9.1, ArchiMate® 2 Certified (level 1 & 2) v2.1 and IT4IT™ Foundation (level 1) from Good e-Learning are Accredited Training Courses and comply with the accreditation requirements for The Open Group Certification for People: TOGAF®, ArchiMate®, and IT4IT™ Certification Programs. IT4IT™, ArchiMate® and TOGAF® are registered trademarks of The Open Group.

Article

Different Approaches to Enterprise Architecture

Svyatoslav Kotusev

Abstract

Enterprise Architecture (EA) is a description of an enterprise from an integrated business and IT perspective intended to bridge the communication gap between business and IT stakeholders and, thereby, improve business and IT alignment. Unfortunately, many companies are dissatisfied with popular heavyweight approaches to EA due to their excessive clumsiness and rigidity. At the same time, alternative lightweight and flexible approaches to EA have also been proposed; however, their existence is not widely acknowledged. In this article I briefly describe these alternative approaches to EA, compare them with the widely-known heavyweight approach, and illustrate their applications in real companies.

Keywords

Enterprise Architecture (EA), problems, approaches, traditional, MIT, DYA

INTRODUCTION

Presently, Information Technology (IT) plays a significant role in improving the competitive positions of many companies. However, many companies investing substantial funds in IT systems do not get the expected returns. Often the effectiveness of IT investments is undermined due to misalignment of business and IT plans. In other words, IT departments often implement information systems that do not address the real strategic needs of their companies.

In order to address the potential misalignment between business and IT strategies the concepts of information systems architecture and then Enterprise Architecture (EA) were introduced (Kotusev 2016). EA is an overarching plan describing organizations from the integrated business and IT viewpoints. EA takes a holistic perspective and shows the relationship between business goals, strategies and processes, and IT capabilities, systems, and technologies. Typically, EA describes business, data, applications, and technology domains and their relationship (TOGAF® 2011). Using EA helps bridge the communication gap between business and IT stakeholders, improve the alignment between business and IT plans, and, thereby, increase the returns on IT investments. Presently, the potential of EA as a powerful organizational instrument is widely recognized and the majority of large companies have started EA programs (van der Raadt et al. 2007).

The classical approach to EA was inspired by conventional engineering and architecture methods (Spewak and Hill 1992). Unsurprisingly, it is largely mechanistic, relies on extensive formal documentation, and advocates following strict sequences of steps to

develop and use EA. This traditional heavyweight step-wise approach to EA was highly influential and essentially shaped the modern understanding of EA theory and practice (Spewak & Tiemann 2006). In fact, most modern EA methodologies are derived from this traditional approach. Unsurprisingly, they also recommend developing a considerable volume of EA documentation and suggest approximately the same sequence of steps to practice EA (Bernard 2012; FEAF 1999; TOGAF 2011).

Many companies willing to improve their business and IT alignment embark on EA journeys following popular formal and heavyweight approaches to EA such as the TOGAF Architecture Development Method (ADM) (TOGAF 2011). Unfortunately, many of these companies fail to succeed with EA because of the excessive rigidity and clumsiness of selected approaches (Holst & Steensen 2011; Lohe & Legner 2014). At the same time, alternative, more lightweight and flexible approaches to EA had also been proposed (Ross et al. 2006; Wagter et al. 2005). These alternative approaches can help EA practitioners struggling with the popular heavyweight documentation-oriented approaches to master EA and achieve better alignment. However, the alternative approaches were undeservedly left unnoticed by EA practitioners and academics and even the very existence of different approaches to EA is still not widely recognized (Kotusev et al. 2015). In this article I will discuss the alternative lightweight approaches to EA, compare them with the traditional heavyweight approach, and illustrate these approaches at work in real companies practicing EA.

TRADITIONAL APPROACH TO EA

The traditional approach to EA, as a revamped version of the earlier IBM's Business Systems Planning (BSP) approach (Kotusev 2016), was initially presented by Spewak and Hill (1992) and subsequently inspired most modern EA methodologies (Armour et al. 1999; Bernard 2012; FEAF 1999; TOGAF 2011). These methodologies suggest that EA is developed step-by-step by enterprise architects who are competent in both business and IT, and possess good communication and systems thinking skills. First, enterprise architects study the current business processes and IT landscape and then document them in detail with a large number of models, diagrams, or blueprints. Second, enterprise architects develop the desired long-term future state according to the strategic plans and objectives of the organization's executives. Typically, the future state development starts from the business architecture as the main driver of EA efforts and then proceeds to the data, applications, and technology architectures. Third, enterprise architects analyze the gaps between the current and future states and develop the transition plan describing the information systems that should be implemented to transform the organization into the envisioned target state from the current position. After being developed, this transition plan is used by the project teams implementing the necessary IT systems under the supervision of enterprise architects. When the plan is implemented, a new EA project or iteration is initiated to carry out the same sequence of steps all over again. In the course of this process, business stakeholders are also expected to use EA documentation for analysis and decision-making.

The traditional approach to EA is shown schematically in Figure 1.

The advantage of the traditional approach is its conceptual simplicity and intuitive appeal. EA planning for the whole organization is carried out in a centralized manner by a small dedicated team and does not require significant involvement of other stakeholders. Since most EA activities are accomplished by a group of expert enterprise architects, it is not necessary for an organization to be particularly IT-savvy to use this approach. Therefore, the traditional approach to EA is best applied at small, centralized, and stable organizations, especially if their competitive advantages are not very dependent on IT.

A successful case of the traditional approach to EA is illustrated in Vignette 1 (overleaf).

PROBLEMS OF THE TRADITIONAL APPROACH

The traditional approach to EA is, undoubtedly, the most discussed and widely known approach to EA. Essentially, it defines the present understanding of EA

practice as well as the understanding of the very notion of EA. Unsurprisingly, the concept of EA and traditional approach to EA are synonymous for many practitioners.

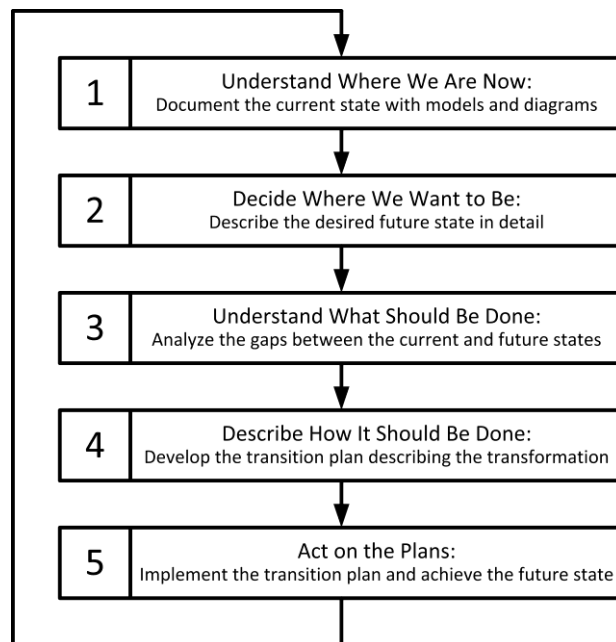


Figure 1: The Traditional Approach to EA

However, despite its popularity in EA literature, the traditional approach to EA is heavily criticized. EA practitioners disparage the traditional approach for its excessive clumsiness because it prescribes following overly complicated processes and creating an unreasonable number of descriptive models (Lagerstrom et al. 2011). EA efforts often fail and many companies become disillusioned with EA altogether because of the overly mechanistic focus of the traditional EA methodologies (Holst & Steensen 2011).

"The problem here is that the enterprise isn't an ordinary system like a machine or a building, and can't be architected or engineered as such." (Bloomberg 2014, p.1).

Lohe and Legner (2014) identify three typical problems with the traditional approach to EA and all of them result from its mechanistic, documentation-oriented, heavyweight, and clumsy nature:

1. Unreasonable efforts needed to create and maintain the necessary EA documentation due to high organizational complexity, large scope, and dynamic environment
2. Low utilization of EA documentation due to its poor quality, obsolescence, wrong level of detail, and mismatch with the real information needs of its stakeholders
3. Poor integration of EA practice into the organization due to its isolated step-wise lifecycle

These problems make the traditional approach unsuitable for large, dynamic, and decentralized organizations working in volatile environments and requiring significant involvement of various stakeholders in decision-making processes.

An unsuccessful case of the traditional approach to EA is illustrated in Vignette 2.

Successful traditional approach to EA at Australian Bureau of Statistics (Gregor et al. 2007; Lynch 2006)

Australian Bureau of Statistics (ABS) is one of the most admired statistical agencies in the world. It is a relatively small and stable government organization operating since 1905. Currently ABS employs more than 3,000 people and provides a high-quality statistics in two major areas: economic and population.

The EA program at ABS was initiated in 1999 when the ABS executive formed a small and multi-disciplinary architecture team of four people responsible for EA development. The architecture team works according to the home-grown EA method based on the traditional EA methodologies and frameworks. Therefore, the ABS approach to EA includes documenting the current state, developing the target state, analyzing the gaps, and then identifying the necessary IT investments to close these gaps.

The architecture team completed the first version of the necessary EA documentation in September 2001 and then updated it to version 2.0 in March 2003. Subsequently, EA documentation is reviewed and updated on a periodical basis every 6-12 months by the technical leadership team headed by the CIO. EA documentation describes ABS at three abstraction levels: business, logical, and physical. After being developed or updated, EA documentation is communicated to all ABS employees with a number of A3 size posters demonstrating the main architectural diagrams. The ABS EA Director argues that EA is “a corporate artifact to be ‘taken away and used’ by all staff at ABS”. Thereby, EA guides the activities of multiple business and IT stakeholders at ABS.

Using EA helps ABS to achieve a high degree of alignment between business objectives and organizational information systems. Unsurprisingly, EA practice at ABS is highly respected by other government organizations. However, most Australian agencies could not replicate the ABS success with EA due to their larger scopes, more dynamic businesses, and decentralized environments.

Vignette 1: Successful Traditional Approach to EA

Unsuccessful traditional approach to EA at the US Federal Government (Gaver 2010)

In 1996 US Congress had enacted the Clinger-Cohen Act obliging the Federal Government and all its departments to develop consistent architectures in order to improve the usage of information systems in this large and decentralized organization. As a response, in 1999 the Federal CIO Council initiated the Federal Enterprise Architecture (FEA) program and published the corresponding FEA Framework (FEAF). FEAF suggested describing business, data, applications, and technology architectures in a segmented manner and embodied the traditional approach to EA including documenting current and future states, analyzing gaps, developing transition plans, and implementing them.

However, the FEA program has faced numerous challenges since its inception. The huge efforts required to create EA documentation undermined the institutional commitment and lowered the priority of EA-related activities. A lack of common understanding of what should be included in EA and terminological issues in FEAF caused disputes and misunderstanding among various program participants. Manual modeling was a very slow process and EA documents often got outdated before being delivered. Most managers who were expected to work with EA documentation could not understand it, especially if they were unfamiliar with modeling.

Consequently, EA diagrams were unsuitable for analysis and reporting and the data from architecture repositories was almost never used. Producing EA documentation quickly turned into an end in itself. EA-related activities hindered normal IT activities instead of facilitating them. Periodical EA maturity assessments showed that the overall EA maturity of the Federal Government was decreasing rather than increasing over the years.

Therefore, despite the best efforts of the architects involved in the program, FEA produced only minimal value for the money spent and never accomplished its original goals. Taking into account that the total expenditures for the program exceeded a billion dollars and much of it (perhaps even most of it) had been wasted, FEA is arguably the most impressive failure of the traditional approach to EA. “*Look at all the efforts that have been launched under the idea of architecture and all the money that has been spent under the umbrella of architecture that has all resulted in unusable shelfware*”, commented Paul Brubaker, co-author of the Clinger-Cohen Act.

Vignette 2: Unsuccessful Traditional Approach to EA

ALTERNATIVE APPROACHES TO EA

As a reaction to the aforementioned problems of the traditional approach to EA caused by its clumsiness and sluggishness, two alternative, more pragmatic approaches to EA had been proposed by Ross et al. (2006) and Wagter et al. (2005). These approaches are more flexible and can help overcome the typical challenges associated with the heavyweight traditional approach. However, they were undeservedly left unnoticed by EA practitioners and academics (Kotusev et al. 2015).

I will discuss them further under the titles MIT (Ross et al. 2006) (because it was developed at MIT) and DYA (Wagter et al. 2005) (because this title is given by its authors) since they do not have any established titles in EA literature.

MIT Approach

The MIT approach to EA was derived from research findings at Massachusetts Institute of Technology (MIT) Sloan School of Management by Ross et al. (2006). Ross et al. (2006, p.vii) argue about the historic ineffectiveness of the traditional approaches to EA and criticize them for:

“... remoteness from the reality of the business and their heavy reliance on mind-numbing detail represented in charts that look more like circuit diagrams than business descriptions and that are useful as little more than doorstops.”

According to the MIT approach, as a first step of EA program business and IT executives should decide on the organizational operating model, defined as:

“the necessary level of business process integration and standardization for delivering goods and services to customers”

Ross et al. (2006) argue that an operating model provides a more clear, actionable, and stable basis for EA development than a business strategy. As a second step, business and IT executives should collaboratively develop the core diagram – a critical EA document describing the main business and IT capabilities, corporate data, principal customers, and key technologies. The core diagram is a one-pager that represents a long-term abstract enterprise-level architectural vision reflecting the integration and standardization requirements of the chosen operating model. Finally, business and IT executives should design and implement the IT engagement model, defined as:

“the system of governance mechanisms assuring that business and IT projects achieve both local and company-wide objectives”

The IT engagement model includes three essential elements:

1. Enterprise-level IT governance – top management decision-making framework including the core diagram
2. Project management – disciplined project delivery methodology with necessary checkpoints
3. Linking mechanisms – processes and committees ensuring the connection between enterprise-level decisions and project-level activities

After the IT engagement model is established, EA embodied in the core diagram is continuously translated into concrete project-level decisions through the linking mechanisms involving business and IT managers on different organizational levels. Therefore, in the MIT approach each IT project eventually accomplishes both local and global objectives and gradually moves a company towards the desired long-term architectural vision.

The MIT approach to EA is shown schematically in Figure 2.

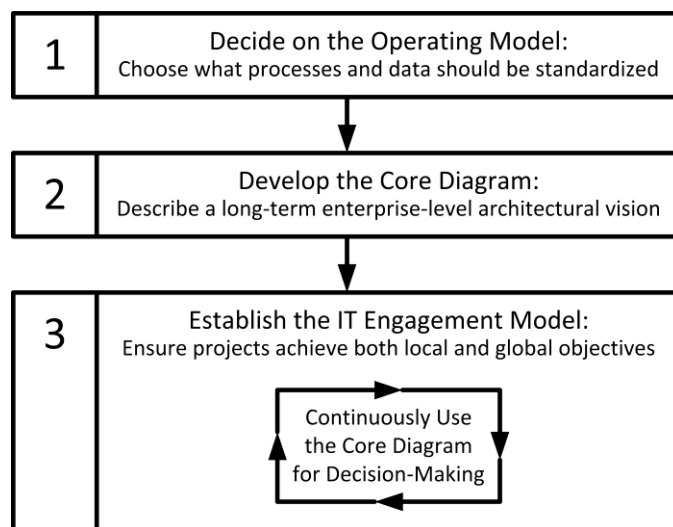


Figure 2: The MIT Approach to EA

The MIT approach to EA facilitates the balance between enterprise-wide and local needs. It provides an organization with a long-term global architectural vision but leaves enough flexibility to react to emerging short-term local requirements. However, the MIT approach requires a very considerable involvement of business stakeholders with an EA program and the establishment of complex architecture governance mechanisms permeating the whole organization. Therefore, the MIT approach is best applied at large and complex organizations with relatively stable business models heavily dependent on IT for achieving competitive advantages in their markets.

A successful case of the MIT approach to EA is illustrated in Vignette 3.

DYA Approach

The DYA (DYnamic Architecture) approach to EA was developed at Sogeti Nederland in 2001 and presented internationally by Wagter et al. (2005). Wagter et al. (2005) criticize the traditional approach to EA for its impracticality and notice that following the recommended processes and developing the heaps of recommended documentation often result in useless “paper tigers” instead of working architecture.

The DYA approach advocates “just enough, just in time” architecture; no EA is designed until there is a need for it. All EA activities in the DYA approach are carried out by architectural services and always triggered by concrete business initiatives emerging in the process of the strategic dialogue when business and IT managers collaboratively decide on which objectives to pursue. After business and IT leaders have determined the potential business initiatives to be implemented, they are elaborated into concrete business cases and project proposals and the role of architectural services in this process is to provide necessary principles and models to facilitate decision-making, impact analysis, and financial analysis. Then, architectural services prepare a project-start architecture describing the scope, design choices, standards and guidelines for the new IT project in order to ensure that this project seamlessly fits into the existing IT landscape and larger picture. Development teams typically use provided project-start architectures in their projects (development with architecture). However, sometimes projects are implemented without project-start architectures (development without architecture) if there are justifiable reasons for it; for instance, acute time pressure, legacy systems involved, or necessary resources unavailable.

The DYA approach to EA is shown schematically in Figure 3.

The advantage of the DYA approach is its totally pragmatic “just enough, just in time” attitude towards EA. It provides the right stakeholders with the right EA documentation at the right time for the right purpose. The DYA approach focuses only on concrete business initiatives instead of abstract intangible visions emphasizing utilitarianism, agility, and flexibility. Naturally, the DYA approach does not imply any long-term strategic planning beyond individual business initiatives potentially undermining the general strategic effectiveness. Therefore, the DYA approach is best applied at organizations operating in very vibrant, dynamic, and unpredictable environments and markets.

A successful case of the DYA approach to EA is illustrated in Vignette 4.

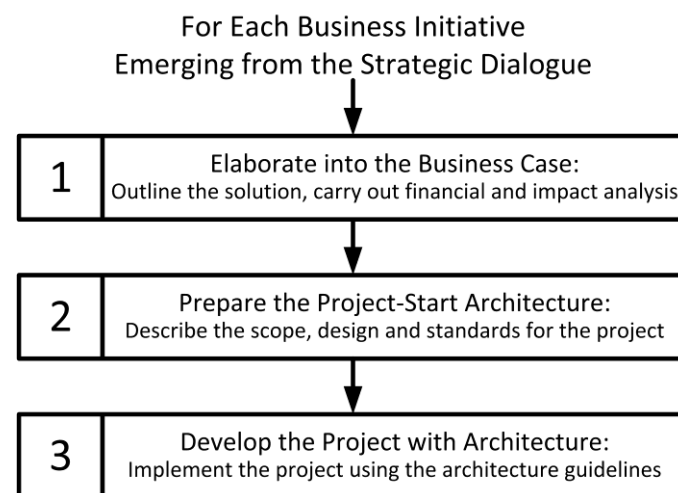


Figure 3: The DYA Approach to EA

COMPARISON OF THE THREE APPROACHES TO EA

Each of the three approaches discussed employs EA to facilitate business and IT alignment. However, each of them employs it in a different manner, follows different logic, and even the very nature of EA is different in each of them (Kotusev et al. 2015). Unsurprisingly, each approach has its own advantages, disadvantages, and situations when it is best applied.

Comparison of the three approaches to EA is summarized in Table 1.

CONCLUSION

EA is an important organizational instrument employed by many large companies to improve the quality of their IT landscapes and align them with the business goals and strategies. Traditionally, EA is associated with a large number of models and complicated processes. However, heavyweight and documentation-oriented EA methodologies often fail due to their excess rigidity. On the other hand, alternative, more flexible and lightweight EA methodologies have also been proposed but their existence is, arguably, not recognized by the majority of EA practitioners and academics. In this article I presented a broad overview of existing EA methodologies, briefly described them, illustrated their work on real-world examples of EA practice, and compared them with each other. Since this article provides only a brief review of EA methodologies and covers only a tip of the EA iceberg, I encourage the readers to study the different approaches to EA in detail using their definitive publications (Ross et al. 2006; TOGAF 2011; Wagter et al. 2005).

Successful MIT approach to EA at Delta Air Lines (Ross et al. 2006; Weill & Ross 2004)

Delta Air Lines, established in 1929, is one of the world's largest airlines. Currently Delta employs more than 77,000 people and serves an extensive US and international network including more than 300 destinations on six continents.

The EA program at Delta started in 1997 when its newly appointed CEO found that organizational information systems were developed independently by different functional units and isolated from each other. EA activities followed the MIT approach and were driven by Delta's CIO who assembled a team of senior executives to define the role of IT in the company. The executive team decided on four standard enterprise-wide processes supported by IT (customer experience, operational pipeline, business reflexes, and employee relationship management) and nine global databases critical to their execution. After that, IT leaders chose the appropriate IT infrastructure to support these processes and data and the resulting high-level architectural vision was fixated in Delta's core diagram. Totally, it took about 60 iterations before all the members of the executive team agreed on the core diagram. This core diagram was subsequently used by Delta's governance bodies for decision-making to allocate IT investments and prioritize projects. Combining the simultaneous investments in applications and underlying infrastructure helped balance short-term and long-term needs and eventually build a solid platform for implementing new business initiatives in a timely manner.

In a few years after the commencement of EA program, Delta had significantly improved its financial results and moved from last to first on main industry indicators; for instance, on-time departures, customer satisfaction, and lost baggage.

Vignette 3: Successful MIT Approach to EA

Successful DYA approach to EA at BNM Banking Group

BNM Banking Group (fictional name due to anonymity requirements) is a large international financial institution with multi-billion dollar revenues. BNM employs more than 26,000 people and is among the top 100 largest banks in the world. It operates in a very dynamic market in the Asia-Pacific (APAC) region and provides a full spectrum of financial services to individual and institutional customers.

BNM started a full-scale EA program in 2004 with the help of consultants and then fine-tuned the EA methodology according to its specific needs and environment. The EA function at BNM works in the logic of the DYA approach and produces three main deliverables: blueprints, Solution Architecture Documents (SADs), and High-Level Designs (HLDs). Blueprints are abstract deliverables intended for executive-level stakeholders (board of directors, executive committee, strategy team, etc.). SADs and HLDs are more concrete and technical deliverables intended for lower-level stakeholders (business and IT operations, project managers, project teams, etc.). At first, senior business and IT executives try to define potential initiatives to be implemented in line with the general strategic direction of the bank. Then, the EA function develops a blueprint for each initiative providing the initial assessment of its value, cost, solution, impact, and risks to facilitate informed decision-making. After the initiative has been approved by all top-level stakeholders, the EA function develops SADs outlining its conceptual implementation and then HLDs describing its detailed technical implementation. Finally, EA documentation is passed to program and project managers driving the actual implementation of the initiative according to the planned architecture.

EA at BNM provides the effective instruments to align IT capabilities and business needs, coordinate change, facilitate simplicity and re-usability, consolidate and standardize the IT landscape, speed up the delivery of new IT systems, and gradually decommission the legacy ones.

Vignette 4: Successful DYA Approach to EA

Approach to EA	Traditional	MIT	DYA
Definitive source(s)	Spewak & Hill (1992), Bernard (2012), TOGAF (2011), and other sources	Ross et al. (2006)	Wagter et al. (2005)
The essence of approach	Document current state, describe future state, analyze gaps, develop transition plan, and implement it	Decide on operating model, develop core diagram, and then use it for decision-making at all organizational levels to balance local and global needs	For each business initiative prepare diagrams for decision-making and then project-start architectures for implementation
Essential EA artifacts	Detailed current state, detailed future state, transition plan	Core diagram (abstract enterprise-wide architectural vision)	Principles and policies for the whole organization, supporting documents for individual initiatives

Approach to EA	Traditional	MIT	DYA
Key terms	Current/future (as-is/to-be, baseline/target) state, gap analysis, transition plan (roadmap), iteration	Operating model, core diagram, IT engagement model, linking mechanisms	Strategic dialogue, architectural services, development with(out) architecture, project-start architecture
Advantages	Conceptually simple, does not require significant involvement of stakeholders, organization can be not IT-savvy	Provides a long-term global architectural vision but leaves enough flexibility to react on emerging short-term local requirements	Pragmatic, flexible, and agile, provides only necessary documents for the right stakeholders at the right time for the right purpose
Disadvantages	Scope can be too large, plans can be unstable, documentation can be incomprehensible, isolated from other organizational activities	Requires considerable involvement of business stakeholders and establishment of complex governance mechanisms	Does not imply any long-term strategic planning beyond individual business initiatives, can undermine general strategic effectiveness
Applicability	Small, centralized, and stable organizations where competitive advantages are not very dependent on IT	Large and complex organizations with relatively stable business models dependent on IT for achieving competitive advantages	Organizations operating in very vibrant, dynamic, and unpredictable environments and markets

Table 1: Comparison of the Three Approaches to EA

ABOUT THE AUTHOR

Svyatoslav Kotusev is a researcher at RMIT University, Melbourne, Australia. He has spent the last three years studying EA practices in organizations. Prior to his academic career he held various software development and architecture positions in industry. He is a TOGAF® Certified expert. Svyatoslav can be reached at kotusev@kotusev.com.

REFERENCES

F.J. Armour, S.H. Kaisler, S.Y. Liu: Building an Enterprise Architecture Step by Step, IT Professional (1:4), pp.31-39 (1999).

S.A. Bernard: An Introduction to Enterprise Architecture (3rd Ed.), Bloomington, IN: AuthorHouse (2012).

J. Bloomberg: Is Enterprise Architecture Completely Broken?, Forbes (2014); retrieved November 11, 2014 from: www.forbes.com/sites/jasonbloomberg/2014/07/11/is-enterprise-architecture-completely-broken/.

FEAF: Federal Enterprise Architecture Framework, Version 1.1, Chief Information Officer Council, Springfield, VA (1999).

S.B. Gaver: Why Doesn't the Federal Enterprise Architecture Work?, Technology Matters, McLean, VA (2010).

S. Gregor, D. Hart, N. Martin: Enterprise Architectures: Enablers of Business Strategy and IS/IT Alignment in Government, Information Technology & People (20:2), pp.96-120 (2007).

M.S. Holst, T.W. Steensen: The Successful Enterprise Architecture Effort, Journal of Enterprise Architecture (7:4), pp.16-22 (2011).

S. Kotusev: The History of Enterprise Architecture: An Evidence-Based Review, Journal of Enterprise Architecture (12:1), pp.29-37 (2016).

S. Kotusev, M. Singh, I. Storey: Consolidating Enterprise Architecture Management Research, in Proceedings of the 48th Hawaii International Conference on System Sciences, R.H. Sprague (Ed.), Kauai, HI: IEEE, pp.4069-4078 (2015).

R. Lagerstrom, T. Sommestad, M. Buschle, M. Ekstedt: Enterprise Architecture Management's Impact on Information Technology Success, in Proceedings of the 44th Hawaii International Conference on System Sciences, R.H. Sprague (Ed.), Kauai, HI: IEEE, pp.1-10 (2011).

J. Lohe, C. Legner: Overcoming Implementation Challenges in Enterprise Architecture Management: A Design Theory for Architecture-Driven IT Management (ADRIMA), Information Systems and e-Business Management (12:1), pp.101-137 (2014).

N. Lynch: Enterprise Architecture – How Does it Work in the Australian Bureau of Statistics?, in Proceedings of the 17th Australasian Conference on Information Systems, S. Spencer, A. Jenkins (Eds.), Adelaide, Australia: Association for Information Systems, pp.1-14 (2006).

J.W. Ross, P. Weill, D.C. Robertson: Enterprise Architecture as Strategy: Creating a Foundation for Business Execution, Boston, MA: Harvard Business School Press (2006).

S.H. Spewak, S.C. Hill: Enterprise Architecture Planning: Developing a Blueprint for Data, Applications, and Technology, New York, NY: Wiley (1992).

S.H. Spewak, M. Tiemann: Updating the Enterprise Architecture Planning Model, Journal of Enterprise Architecture (2:2), pp.11-19 (2006).

TOGAF® Version 9.1, an Open Group Standard; refer to: www.opengroup.org/togaf.

B. van der Raadt, R. Slot, H. van Vliet: Experience Report: Assessing a Global Financial Services Company on its Enterprise Architecture Effectiveness Using NAOMI, in Proceedings of the 40th Hawaii International Conference on System Sciences, R.H. Sprague (Ed.), Big Island, HI: IEEE, pp.1-10 (2007).

R. Wagter, M. van den Berg, J. Luijpers, M. van Steenbergen: Dynamic Enterprise Architecture: How to Make it Work, Hoboken, NJ: Wiley (2005).

P. Weill, J.W. Ross: IT Governance: How Top Performers Manage IT Decision Rights for Superior Results, Boston, MA: Harvard Business School Press (2004).



Making Standards Work®

The Open Group Events 2017

**The Open Group Berlin Event
and Member Meeting**
Making Standards Work®, Smart Cities
Berlin, Germany
April 24 - 27, 2017

**The Open Group Manila and
AEA Philippines**
Using Enterprise Architecture for
Business Transformation
March 16 - 17, 2017

If you are a member of the AEA you can receive a 10% reduction on the non-member rate by using the following promotion codes on registration: BER174-AEA for Berlin and OTT177-AEA for Ottawa.

**The Open Group Ottawa Event
and Member Meeting**
Making Standards Work®, e-Government
Ottawa, Canada
July 17 - 20, 2017

**The Open Group Amsterdam Event
and Member Meeting**
Making Standards Work®,
Architecting the Digital Enterprise
October 23 - 26, 2017

A place to exchange ideas and build relationships beneficial to your organization

The Open Group events provide a venue to gain valuable knowledge, and participate in Forums and Work Groups that are developing the next generation of open, vendor-neutral standards and certifications. Those who attend The Open Group events benefit from the opportunity to leverage the expertise of other experts, learn from others' experiences, and explore content that is relevant to their jobs and organizations.

Events also present an opportunity for attendees to access a wide range of experts, practitioners and specialists in a non-sales environment, and build professional relationships beneficial to you and your organization. The Open Group hosts quarterly events, as well as regional events hosted by our local partners around the globe.



Our Platinum Members



Article

Theoretical Perspectives of Enterprise Architecture for Technological Transformation

Torben Tambo

Abstract

The purpose of this article is to investigate the completeness of the theoretical foundations of Enterprise Architecture (EA) by reviewing four selected disciplines from Management of Technology (MOT). Often theory on EA is based on prior EA contributions or more distant contributions such as service science, semiotics, psycho-social constructs, business process analytics, and systems science. It is here argued that other theories might be more supportive to EA. The current article is based on a review of the MOT literature and a subsequent literature review within each of the four specialized disciplines. Furthermore, the article includes two qualitative, longitudinal case studies mapped onto the four disciplines. It is demonstrated that EA can benefit from theoretical positions closer to EA than usually selected. It is furthermore demonstrated that innovation and technology management significantly overlap EA, for which reason they could eventually be amalgamated. A model view is proposed where activities of EA are augmented with the four theoretical positions. EA is occasionally self-referential in its theoretical argument. As a research implication, this article suggests aligning EA with closely-related activities of the enterprise. EA is often at risk in the enterprise to be retrospective towards changes. The suggested framework has practical implications in order to organize EA activities more collaboratively and cross-functionally; e.g., include marketing officers and librarians in EA teams. This article claims originality in its proposition to align EA with a multi-disciplinary approach derived from MOT.

Keywords

Enterprise Architecture, Innovation Management, Technology Intelligence, Technology Roadmapping, Knowledge Management, Management of Technology, MOT

INTRODUCTION

Enterprise Architecture (EA) is typically seen as a transformation regime directly attached to or derived from business strategy (Ross et al. 2006; Ross et al. 2003). Theoretical models of EA often draw links between the technological subsystems of the company and the strategic management elements (Tamm et al. 2011). In this article, it is assumed that the strategy-technology link can be improved from a theoretical point of view by considering business-driven technological transformations more broadly. Discussing and developing EA is often based on either empirical investigations (Närman et al. 2013) or formal or industrial frameworks (Fischer et al. 2007; Braun & Winter 2005). These discussions and knowledge disseminations are important to ensure professional learning and methodological rigor. However, more fundamental explanatory theoretical approaches can enrich insight and avoid self-referential positions. The extension of the theoretical view thus contributes to a more meta-theoretical approach to EA.

The strategy-technology axis can typically not stand alone. Strategy must be implemented through a host of

guidelines and policies (Magoulas et al. 2012). Strategy is often short-lived and operational practices can greatly impact EA (Lankhorst 2004). Knowledge elements are found throughout the enterprise, and EA relies on detailed insight into business processes and specific knowledge of the enterprise and its systems (Donate & Sanchez de Pablo 2014; von Krogh & Geilinger 2014). EA is time-dependent and has the purpose of organizing planning based on expected changes in the future. EA is also about conducting necessary technological changes in the enterprise; often EA is idealized as related to new businesses, but much EA has to reflect incremental changes found broadly within and outside of the enterprise as well as in regular lifecycle management of technologies.

Several EA contributions present meta-theoretical positions on social or individual behavioral aspects of EA; e.g., the sense-making theory of Weick (Møller et al. 2012; Joubert 2010; Kahn 2012) and the business process management and modeling approaches of Davenport (Tamm et al. 2011; Ross 2003). Business ontologies as well as interpretive and translation theories are also widely used (Kang et al. 2010). Several of such theoretical approaches can be too fundamental and may

tend to separate themselves from practice. This article suggests identifying theories that address continuous technological change throughout the enterprise.

In introducing meta-theoretical elements in an EA discussion, it is worth considering relevance, purpose, and impact. Here, technology and technological change are in focus, both in terms of products related to markets and the impact of this to the company, but also in relation to internal process changes and improvements requiring attention. In the following, theoretical positions are introduced with inspiration from the engineering discipline of Management of Technology (MOT). Four elements of immediate interest are pinpointed (Schilling 2012, Badawy 2010; Badawy 1995; Takeuchi & Nonaka 1986):

- Innovation management
- Technology intelligence
- Technology roadmapping
- Knowledge management

Through the remainder of this article, the theoretical constructs related to EA will be introduced and discussed in order to evaluate the applicability and value within EA.

The current article is methodologically based on a literature review and two longitudinal case studies (Bryman & Bell 2007). The methodology is inspired by the interpretive and social thinking adopting a position like Mathiassen and Nielsen (2008) and Walsham (1998), and is predominantly qualitative.

The cases have been compiled during 2012 to 2014 using interviews with senior management, mapping of system architectures, and open sources, such as websites, press releases, advertisements, and supplier information. The cases are to a certain degree heterogeneous (Pettigrew et al. 2001), but they are neither extreme nor polar types.

This article is based on the information system research tradition of observing and understanding (information) technology within its context in organization, application, and business (Mathiassen & Nielsen 2008; Kock 2007). An account of (lack of) technological insight of information systems research is found in Koch et al. (2010). MOT is viewed as a multi-paradigmatic approach applied to study objects of the organizational context at both management and operational levels (Schultz & Hatch 1996).

THEORY

The origin of EA is subject to debate, but Zachmann (1987) is widely recognized as the “founding father” of EA (Noran 2003). The interest in the business-technology alignment must be seen in the light of the

narrow technical focus of the information technology professionals at that time. Since then, a number of different “schools” have originated (Lapalme 2012), some with a stronger focus on technology and “quantitative” methods, others with a distinct organizational and process-oriented focus (Bernus 2003). Bernard (2012) describes EA as a transformation agenda where both current state and future state of the architecture must be created on an enlightened basis and with a strong adherence to business requirements.

Bouwman et al. (2011) demonstrate an inter-relation between technological EA and a non-technological conceptualization of the business process referred to as business architecture derived from process flows, legal procedures, norms, and values. The use of values is further elaborated in Korhonen & Joutanen (2013) who define EA as derived from three “meta-physical” architectures: technical, socio-technical, and eco-systemic.

EA covers intra-organizational areas of technology, but also inter-organizational aspects (Garg et al. 2006; Chandra & Kumar 2001). The technological dimension of EA ranges from fundamental infrastructural components to the detailed level of business system support and work processes with associated mapping tools (Pulkkinen 2006; Taleb & Cherkaoui 2012). EA is applied in commercial contexts as well as governmental sectors (Hjort-Madsen 2009; Jensen 2012) where the latter can be early adopters based on required or recommended EA practices as part of risk mitigation and political decision-making (Afshani et al. 2012).

Innovation Management

Innovation management is a broad and cross-disciplinary set of governance approaches for any sort of technologically-based corporate change, although mostly focusing on the organization’s ability to create new products (Schilling 2012; Miles & Snow 2007; Nill & Kemp 2009). Despite its technological connotations, innovation management tends to be an economic discipline based on the fact that most companies’ revenue streams will change significantly over time and only proper innovation management can ensure that revenue streams drying out are replaced by new sources of income (Hervas-Oliver & Sempere-Ripoll 2014; Brem & Voigt 2009; Garcia & Calantone 2002; Etzkowitz & Leydesdorff 2000). The Organization for Economic Co-operation and Development (OECD) defines innovation as one of three fundamental types: product innovation, process innovation, and organizational innovation (OECD 2005). By relating itself to Information Technology (IT), EA tends to limit itself to process innovation, even if many contemporary product innovations rely strongly on IT.

Van de Ven (1986) suggests innovation as the responsibility of the organization, but this includes circumstances where individuals can maintain attention, learn from relationships, especially with customers, and collaborate in environments in which much is shared:

“Innovation has been defined as the development and implementation of new ideas by people who over time engage in transactions with others within an institutional order.”

Here, organizational approaches are needed to process ideas, understand critical limits, conduct experiments, select, learn, improve, and manage uncertainty (Crossan & Apaydin 2010).

Bergek et al. (2009) argue that innovation is not isolated, but can only take place in an organizational innovation system or a Technological Innovation System (TIS) which furthermore must ensure knowledge development and diffusion, research direction, “entrepreneurial experimentation”, market formation, legitimization, and resource mobilization (Kapsali 2011; Johannesen et al. 1997). Sundbo & Gallouj (1998) are sceptical about the “well-organized” innovation management, but describe patterns of innovation, particularly within service industries, as less well-organized compared to manufacturing industries. Desouza et al. (2009) suggest that innovation management should focus on the following issues: idea generation and mobilization, screening and advocacy, experimentation, commercialization, as well as diffusion and implementation along with extensive systems of success indicators.

Innovation and insight into successful methods of innovation are strongly linked to learning (Brix 2014; Brown & Wyatt 2010) and exchange of knowledge within an organization, and in the case of open innovation also to external parties (Lazzarotti & Manzini 2009; Veugelers et al. 2010). Innovation is often discussed in terms of new products and products with a high impact on economy and society, but most innovations are continuous innovations that produce smaller changes over time (Van Looy et al. 2005).

Innovation management correlates with EA from its focus on corporate and technological transformations. For many organizations, innovation management is predominantly product-centric, whereas EA is process-centric, and therefore tends to be isolated from the innovation management. With the digital transformation of many industries, the distinction between the product and the process perspective is no longer relevant. EA and innovation management are therefore to be considered as mutually interchangeable, although companies may maintain a more narrow view of the two constructs in terms of either process or products.

Technology Intelligence

Technology intelligence is the organized collection of technological information and knowledge primarily from external sources (Lichtenthaler 2007; Porter 2005; Arman & Foden 2010). In their seminal contribution, Kerr et al. (2006) describe technology intelligence as the exchange between intelligence consumers and intelligence brokers unfolded as a chain of information flow moving from decision-making, intelligence needs, actions, and activities to sources and back. Technology intelligence is one of several methodologies for data collection in expectation of future development; others are known as, for example, technology foresight, competitive intelligence, technology scouting, and institutional foresight (Peter & Jarratt 2013; Battistella 2013; Koch et al. 2008). Technology intelligence describes forms of coordination for retrieval of knowledge, especially structural (hierarchically defined) coordination, informal coordination, and hybrid coordination (Lichtenthaler 2004).

Early data collection is known as scanning, whereas later stages of data collection are known as monitoring. Both scanning and monitoring can be classified as passive or active dependent on the company's involvement in the area of knowledge. Brenner (1996) proposes that technology intelligence must convert into competitive intelligence to prove useful. Brenner (1996) furthermore emphasizes that knowledge often originates as “white” (public, open), “grey” (semi-public, hard to obtain), or “dark” (secrets, intellectual property), and then develops over time and intensity (Kerr et al. 2006). Lee et al. (2013) suggest a close relationship between technology intelligence and other actions related to technological transformation, among other things to understand the core of the technologies and the development of acquired knowledge into a technology strategy.

Technology intelligence is closely related to innovation management (Veugelers et al. 2010) with the information collection process often in the foreground due to the overwhelming amount of information sources present on the Internet, among other things. Veugelers et al. present various tools and principles to make technology intelligence useful; e.g., technology radar using academic literature sources, patent databases, and general Internet search word clusters.

Several articles present case studies of process design improvement using technology intelligence. Behkami & Daim (2012) propose technology intelligence as a means of research strategy development in health care informatics.

Castellanos & Torres (2010) argue for a more complex technology intelligence implementation consisting of:

1. Implementation tools
2. Knowledge generation
3. Formulation and implementation of strategies

Smaller industries can join in associations to overcome the cost of technology intelligence (Phillips 2010), thereby democratizing the benefits of technology intelligence; this is in line with Holmes & Ferrill (2005) who describe an initiative in Singapore, where smaller companies' technology roadmapping processes are supported by the government.

Data collected in the technology intelligence process can be both quantitatively and qualitatively processed to create scenarios and models for future situations. Vanston (2003) suggests five main methodologies or "philosophies": extrapolation, pattern analysis, goal analysis, counter punches, and intuition. Extrapolation is the most quantitative using mathematical methods. Intuition is the most qualitative; however, Delphi surveys can be conducted here for smaller or larger cohorts.

The EA literature includes suggestions on collection of information, but generally not on a very systematic basis nor in an extra-organizational or a community context. Technology intelligence therefore suggests an extension to EA that increases the general technological body of knowledge and enables EA to act better as a predictive tool in the organization. There is obviously a risk that EA may lose focus on business and strategy and be solution-centric instead of ensuring governance and alignment. The risk is assumed balanced by the potentials of having EA performed on an improved basis of insight as well as having the EA transformation processes guided by the organized data collection from technology intelligence.

Technology Roadmapping

Technology roadmaps are often referred to colloquially as "simple" and *ad hoc* lists of what to do at what time (Gersri & Kocaoglu 2007; Geum et al. 2011). At Cambridge University, research in Technology Roadmapping (TRM) has been conducted over the last 15 years to establish a more rigorous discipline of tools and methods (Phaal & Müller 2009; Phaal et al. 2006). Carvalho et al. (2013) present an extensive literature review of 79 key articles on TRM from 1997 to 2011, where the research is "judged" as useful and beneficial, but still reflects an area that needs to mature.

TRM is characterized by purpose and format. Purpose can be planning for product portfolio, strategy, knowledge, capability, integration, programs, and projects. Format can be stored in one or more layers,

using graphs, pictures, tables, bars, and similar means such as Gantt and PERT diagrams, although resources are replaced with assumptions, relations, and dependencies of technologies. A standardized process is suggested, also known as the T-plan (Phaal et al. 2001); roadmapping is based on four workshops with the following foci:

- Market
- Product
- Technology
- Roadmapping

In its general form, TRM balances between technology push and market pull, along with a timeline connecting resources, deliveries, and purpose (Arshed et al. 2012).

Petrack & Echols (2004) elaborate TRM to reflect technology S-curves into roadmaps describing progress from research and development to product and market. More specifically, they present a roadmap structure from Motorola covering both corporate internal roadmaps and external roadmaps exchanged with suppliers, customers, and partners. Roadmaps for technology, products, and IP (intellectual property and knowledge) are developed in parallel and exchanged via a structure of meta-roadmaps.

In most cases, TRM has a strategic importance as it constitutes the "master plan" for new technologies, products, and associated processes. This also reflects the activities of innovation management as described above. Gersri et al. (2009) consider TRM implementation as an organizational process characterized by a dynamic of shifting responsibilities: implementation is assumed to occur as an initiation stage, a development stage, and an integration stage, each reflecting organizational and technological maturity in using TRM.

In considering effectiveness of innovation management, Oliveira & Rozenfeld (2010) suggest to integrate TRM and Portfolio Management (PPM). New Product Development (NPD) teams are then better aligned with market goals; this gives a more holistic understanding, and information processing is reduced as TRM and PPM are integrated. Systematic scenario building is proposed to underpin inter-relations between organization and technology in roadmapping (Lee et al. 2014). Communication and dissemination of the roadmaps must establish credibility throughout the organization (Lee et al., 2012).

TRM is procedurally compatible with the transformation perspectives of EA and, for example, the "management plan" of EA. Again, it can be said that the scope is that MOT can be viewed as fully aligned with EA despite the aforementioned traditional way of observing innovation

as more product-oriented and EA as process-oriented. EA has good opportunities to extend its role by assuming the theoretical contributions of TRM, which also in some contexts is referred to as “enterprise architecture roadmapping”.

Knowledge Management

Knowledge management is the organized form of establishing appropriate and informed intellectual capabilities within the organization and as an organizational capability (Gold et al. 2001). As stated above, knowledge can be retrieved from external sources, but internal perspectives of knowledge management are critical to filter, organize, maintain, store, and find (von Krogh & Geilinger 2014). Particularly the internal creation of knowledge is for most enterprises a matter of survival. Internal innovation processes are targeted at knowledge creation. In such processes, knowledge held by individuals (implicit knowledge) must interact with the collective body of knowledge as explicit knowledge in processes of externalization, socialization, internalization, and combination (Nonaka & Takeuchi 1996). The model described below is used by Hoegl & Schulze (2005) in a presentation of a number of knowledge creation methodologies; e.g., workshop, communities of practice, and expert interviews.

Hung et al. (2008) introduce a number of factors critical to knowledge creation. Mechanisms must match capability. An experimental dimension is assumed as always prevalent, and efficiency, flexibility, scope, and appropriateness of the experimental practice are then amalgamated into knowledge.

Some scholars see technology intelligence as the antecedent of knowledge management (Arman & Foden 2010), where knowledge management will add value to technology intelligence by offering company-internal structures and technologies to support, enrich, and maintain knowledge.

Knowledge management is often described implicitly in relation to EA, where it is obvious that EA is knowledge-intensive and tools at hand aim at providing externalization and explication. The knowledge perspective is generally not clearly stated as an EA activity, but given EA as a dyad between the architect and the domain in exchange of knowledge, a knowledge management perspective can contribute to accuracy of the EA processes.

Model of EA and Management of Technology Elements

As indicated, this article assumes congruence between the four meta-theoretical disciplines of MOT and EA.

The congruence is potentially on a level where EA and MOT in certain corporate environments could be interchangeable or even identical. In the presentations above, the disciplines are presented from key contributions each within their field. A further literature review on commonly available search engines will find that each of the individual disciplines is related to EA on an individual basis. In the following, it is thus the idea of having a multi-disciplinary theoretical approach to EA that will enable the discussion of alignment and potential confluence between EA and MOT.

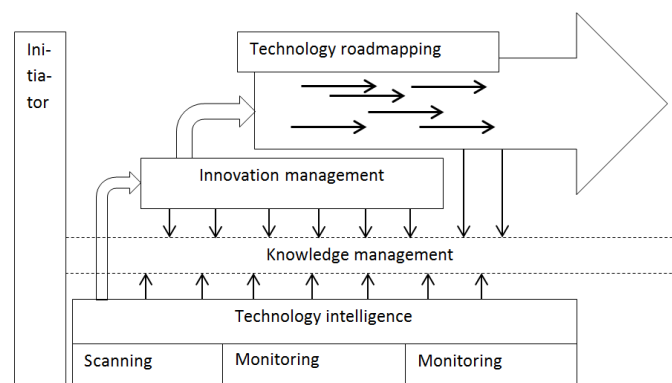


Figure 1: Meta-Theoretical Model of Interaction and Process

Insight acquired through technology intelligence is used to guide innovation management in developing technology roadmaps within all processes relying on knowledge management. Figure 1 omits business strategy, as strategy can be more short-lived than the processes, and at the same time, strategy can both govern the processes, but may also be changed as a result of any of the processes.

CASE STUDIES

In EA research as well as Information Systems (IS) studies, case studies dominate as the principal methodology in gaining empirical insight and testing of impact of changes. In case studies, it is important to exercise uniformity in the applied approach and attempting to aim for what is in fact feasible. In the cases presented below, there are numerous commercial, technological, and organizational aspects working together. However, the cases are contemporary and exemplary in the context of the study of technological transformation.

DigiCorp

DigiCorp is a European provider of digital services with around 6,000 employees. DigiCorp has been in the market for many years, and its systems landscape is complex and diverse. Many systems are 20-30 years old, and several of the systems have somewhat

overlapping scopes; several CRM systems maintain different, not particularly logical groups of customers, and various systems are billing and invoicing customer groups randomly across the CRM system. The following section describes an effort to improve quality within the area of providing services to “simple” private customers.

A private customer would start a purchase by calling a customer service agent. The agent should then register the customer’s interest and issue an order confirmation. During the sales process, the service agent may – via different portals – be in contact with up to 70 different individual IT and technical systems in resolving a customer inquiry. At DigiCorp, most systems were seemingly open with APIs and even SOA enabled. Experience had shown that a high proportion of customer inquiries were answered wrongly. Customers were promised undeliverable services, or they were not offered high-value services that could have been provided. The company wanted to change to reduce costs and give the customers a better service experience. Throughout the years, several vendors had attempted to create the “next generation customer service system” but without luck. Operations of the legacy systems were outsourced, and the company experienced that when a new service provider took over, much knowledge was lost. The market for such services is under heavy competitive pressure and is expected to decline in the next few years.

The company reviewed its options:

1. A standard ERP system was not an option, as a strong, specialized technical integration was required.
2. SOA was by and large implemented, but it became overly complex and required much knowledge to maintain.
3. The range of existing systems would suffer “end-of-life” within a short span of years; i.e., knowledge and support would be non-existent.
4. A different solution combining elements of all options.

The company opted for the fourth option, and went on to establish a joint database covering common technical aspects of the services provided to customers, or services and quality of services that a potential customer would receive. The database should extract data from all relevant technical and administrative systems and contain the master version of the available data. As such, the approach would resemble an ERP system, but it would be much more technically-oriented. The database solution would entail a great deal of redundancy and duplication of data, but would ensure a more speedy customer service process, more uniform data, enable the termination of deprecated systems, and

enable internal or external parties to supply and demand data for given purposes.

Analysis of DigiCorp showed that innovation management to a large extent had failed in terms of the internal process technologies. The company continuously consolidates a number of technical systems, and it also offers new services to its customers. However, only a few of these activities are fully automatically supported by the existing systems, and many new market initiatives are manually supported internally. Technology intelligence is very precise in pinpointing the need for technologies to be consolidated on very few platforms mainly concentrated around one technology. Technology roadmapping is closely related to the business strategy and the technological changes already underway. The market dimension of TRM in DigiCorp changes so rapidly that major revisions are taking place on a quarterly basis. Knowledge is a critical issue; in many cases, DigiCorp has chosen to outsource business units due to the fact that a critical mass of knowledge has been lost, and stronger outsourcing partners do have better chances of accomplishing the task. Knowledge management has therefore largely been outsourced.

From an EA perspective, DigiCorp faces massive changes. Due to the outsourcing of IT operations, IT development, and a range of operational tasks of its service infrastructure, DigiCorp has limited insight into and influence on the processes. However, the decision to create a consolidated database of best possible data is under the given circumstances a major EA decision that will improve customer service and contribute to reduced technological complexity, as many-to-many interfaces are replaced with more one-to-many interfaces.

PayDK

In 2009, the Danish national government decided to establish a national Shared Service Centre (SSC) for case processing and payments of welfare and social benefits based on objective criteria; e.g., child support, maternity support, housing support, and four more services. The motivation was to cut costs from the 98 municipalities that had previously been in charge of this. A semi-governmental service was conceptualized and called “PayDK” (Udbetaling Danmark). In 2010, an implementation organization was set up, and from late 2012 to 2013, services were transferred to PayDK.

PayDK was originally set up as an administrative reorganization and not a technology project. PayDK had limited internal IT resources; however, it had a “master contract” with the dominant IT supplier, KMD, which had previously been owned by the municipalities, but had

been privatized since 2008. The background for establishing PayDK was, besides reducing costs, the need to have a long-term plan for technological and digital development in the governmental sector. In this connection, citizen IDs and portals, where interactions between citizens and the government increasingly were to take place, were highly important. In discussing available channels by which to interact with the government, more and more services are now based on single-channel communication, where all communication for specific services is fully digitized. Digitally divided citizens must look for help from relatives or contact various front-office services; e.g., in public libraries, to use the systems.

During 2011 and 2012, the staff, the services, and the software responsibilities were transferred from the municipalities to PayDK, with the staff organized in five regional centers. Of originally 2,000 employees, 1,000 were to be transferred to PayDK, and 500 were to remain in the municipalities to support citizens who did not have the required digital skills. In the summer of 2013, further assignments were transferred to PayDK in the management of various retirement pensions.

In the architecture of PayDK, it is given that each system will only exist in one version. Over time, each system is assumed to be tendered where KMD and others can bid. 98 individual architectures are thus transferred into one architecture. The void may cause some concern when extracting systems from a running operation. In contrast is a range of positive issues of buying/tendering power, architectural simplification, ensuring single-channel e-government, uniformity in staff training and citizens' experience, and improved fraud detection from a consolidated database.

The transfer of services reflects innovation management from the perspective of ensuring one common, strong citizens service platform and saving the cost of having 98 different platforms with more innovation backlog. Technology intelligence is centralized which at least reduces the need for local insight and effort; in a constant drive towards more e-government, the newest approaches can be collected and analyzed; e.g., the fact that citizens prefer using tablet and smartphone devices to access e-government services rather than classical personal computers and laptops. Technology roadmaps are very clear and include the tendering of former monopolized systems as seen in Figure 2.

Knowledge management is likewise centralized. The impression is that only one of the five regional centers is actually allowed to systematically collect experience from the case processes, business process descriptions, and technological matters.

The project of PayDK represents a major EA transformation in Danish e-government. The transformation covers several facets of EA; e.g., systems consolidation, reorganizing flow of data, organization of a single-channel citizen portal, implementation and user acceptance of a digital identification system, mitigation services for those on the wrong side of the digital divide, and a platform open to new suppliers. The business dimension was said to be the savings coming from 500 fewer employees as well as a more efficient, single instance infrastructure. It is, however, unclear whether the business effects are a result of this, or if they occur from derived effects of, for example, adaptation to single-channel government services, substantial reduction of errors from analog channels, or simplification of municipal IT architecture.

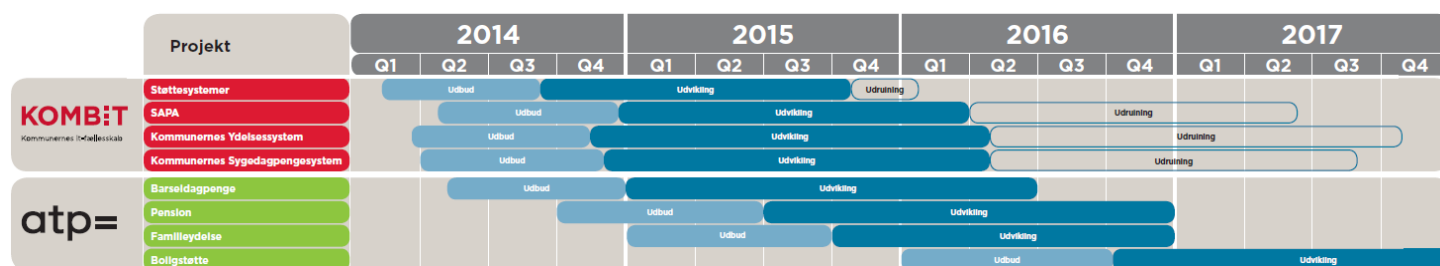


Figure 2: A Technology Roadmap Illustrated as a Tendering Roadmap

	DigiCorp	PayDK	EA Implications
Innovation management	Key projects failed Outsourcing of responsibilities	Unsure Outsourcing, but retaining a core knowledge of processes	Reducing the "precision" of EA Lack of EA insight within the organizations
	Creation of a cross-cutting database	Consolidating of single platforms, but tendering on system level	Re-vitalization of the EA role

	DigiCorp	PayDK	EA Implications
Technology intelligence	Issues of TI when technology operations are largely outsourced	Technological change through tendering	EA associated with outsourcing partners' interaction; appropriate TI critical to successful outsourcing
	Intelligence gathered interacting with current and potential partners	Intelligence gathered in open, cross-organizational forums	Grey and white sources impact EA
Technology roadmapping	Unsure scenarios due to technological change	Specific scenarios from tendering plans	EA roadmaps associated with both stable and volatile organizations
	Database as vehicle for stability and change	Increased interoperability in governmental systems	EA roadmaps support desired business transformations
Knowledge management	Distribution of knowledge management	Clear vendor roles in knowledge management	EA deals with complex and heterogeneous knowledge
	Retention of core business knowledge	Centralization of knowledge management	EA as knowledge management core

Table 1: Meta-Theoretical Case Comparison

DISCUSSION

Is the theoretical foundation of EA sufficient, or would it make sense to extend the theoretical base? Several authors suggest expanding the theoretical scope of EA's systemic character using ontologies and meta-code approaches, while others aim at EA as a governance regime. In this article, we suggest that EA be underpinned with the four meta-theories from MOT, given MOT's close position in the landscape of corporate governance frameworks that are both connected to business strategy and technologically-oriented. In both cases, neither EA nor MOT were explicit drivers for the processes; in fact, each served as tools and supportive mechanisms. Considering DigiCorp's project as a remediation for out-of-date systems and PayDK's stated main purpose of being a human resource project, only high-level-insight in EA is maintained within the organizations, with the main part of EA being outsourced. In Table 1, the cases are compared with respect to the four meta-theories.

From the cases, it is indicated that EA has several common traits with MOT and MOT's underpinning theories, but where are the direct similarities and experiences? Several issues are highlighted in the cases concerning governance, structure, interaction, relation, and organization. Performance measurement of the activities is likewise requested but implicit in the cases. Below (in Table 2) is a set of key commonalities derived from the cases, each describing the challenges and practices in creating organizational structures and

working relations. The term "activities" is used for both EA and the four meta-theories.

The shared commonalities are indicative for potentials and benefits to be gained by aligning or combining the activities. Due considerations must, however, be taken throughout the process that EA and MOT are commensurable in light of EA historically being an incarnation of IT whereas MOT is purely generic in its technology understanding.

To the further discussion, it is still the overall claim that EA and MOT share a high degree of identity, especially when it comes to innovation thinking and management processes. As such, EA meets the criteria for being a Technological Innovation System (TIS) or an organizational innovation system (Bergek et al. 2008; Sundbo & Gallouj 2000). In IT and henceforth EA, open innovation practices are prevalent (Veugelers et al. 2010; Lazzarotti & Manzini 2009), and subsequent innovation in networks, communities, ecosystems, and mutually learning organizations (von Krogh & Geilinger 2014; Battistella 2014; Nill & Kemp 2009; Johannesen et al. 1997). Particularly Miles & Snow (2007) have used the IT industry and the development of IT infrastructure and architecture standards as examples of shared innovation. Like any other innovation, EA ideally needs articulation of demand from leading customers and integration in products and services development (Tamm et al. 2011; Pulkkinen 2006; Chandra & Kumar 2001).

Business strategy	The four meta-theories as well as EA are to have a distinct starting point in business strategy by knowing and supporting strategic objectives (Phaal & Müller 2009; Ross et al. 2006).
Management support	Management support is imperative for funding, but more critical in terms of influence, collaboration at the highest level, and support to gain broader organizational acceptance (Jensen 2010; van de Ven 1986).
Distinct organizational responsibilities	The activities whether EA or MOT must be clearly rooted in a dedicated organizational setting to gain professionalism and momentum. This both increases quality and dedication, but also creates distance and potential alienation to the hosting organization (Hervas-Oliver & Sempere-Ripoll 2014; Bouwman et al. 2011).
Product/process support	Activities not related to the development or improvement of products, services, or relevant processes should not take place. This raises the concerns of having a relatively deep insight into market issues contradict with the specialist approach (Lee et al. 2012; Holmes & Ferill 2005; Tamm et al. 2011).
Embedding of R&D units	The activities should be closely related to the organizational structures of R&D, but the cross-cutting character may also keep a distance. A long distance to R&D can undermine both (Kapsali 2011).
Informal relationship to operating units	As classical staff functions, the organizational relationships are largely informal, thereby relying on personal relations and goodwill. Tentatively, the activities might be retrospective as the core organization is not seeing imminent necessity of engagement in the early stages (Philips 2010).
Longitudinal character	Going across projects, programs, and organizational boundaries is characteristic, thereby functioning as an “organizational memory”.
Measurable (lack of)	Common is the difficulty in creating measurability and quantifiable assessments of the activities especially in comparison to <i>ad hoc</i> , contingent, and emergent alternatives. This creates a common vulnerability and a person-bound dependency (Korhonen & Poutanen 2013; Carvalho et al. 2013).

Table 2: Key Commonalities Derived from the Cases

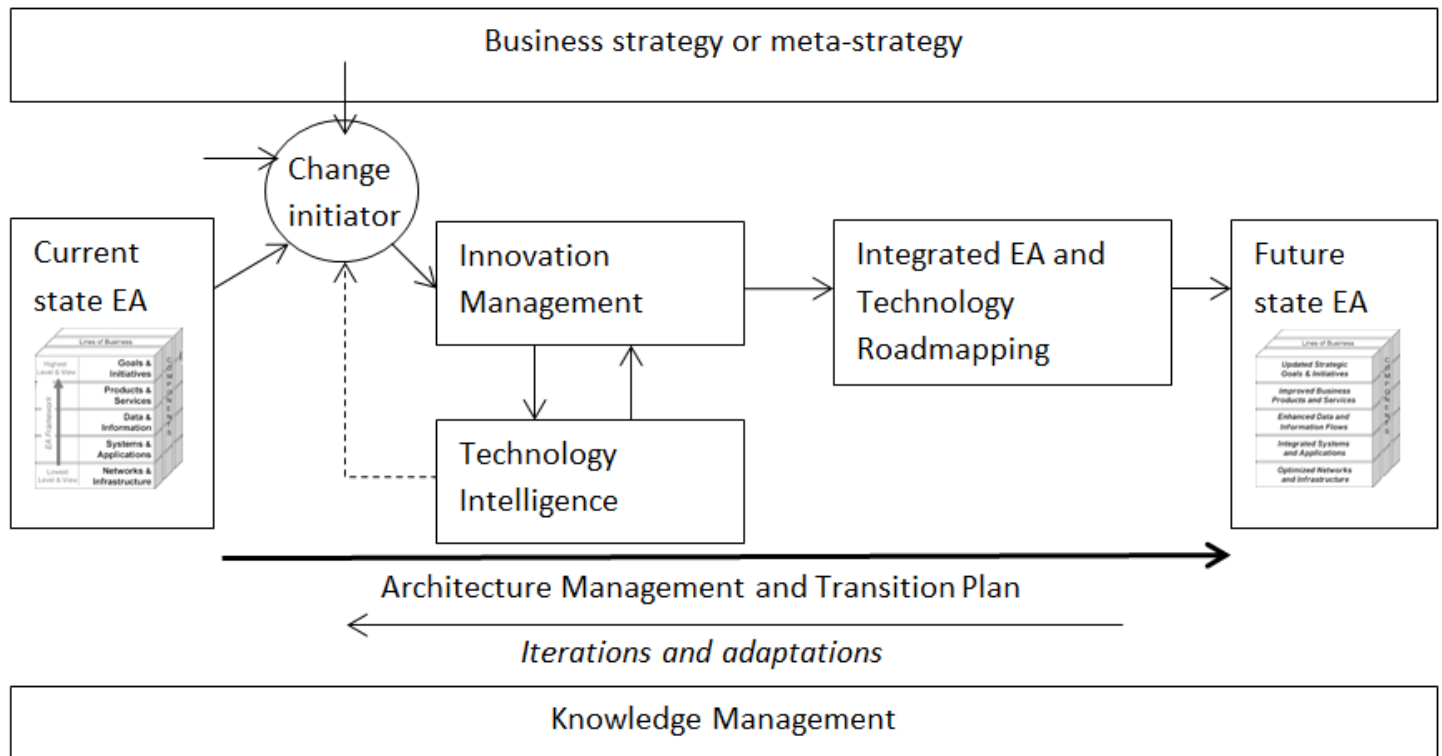


Figure 3: Model for Combined EA and MOT inspired by EA3

Contribution to EA

This article demonstrates that EA runs in parallel with several other processes in the enterprise. The four meta-theoretical approaches can each help to sustain EA, although the cases also cover organizational, strategic, and human issues. The chosen theories of MOT are largely congruent with EA. Although MOT seems to focus more on new product development, it must be maintained that the four theories can relate to both process innovation and retrospective activities. As such, innovation management shares goals with any EA activity related to planning of change. Technology intelligence generally seems to be underestimated in the EA literature as well as in practice. TRM is also fully congruent with EA and by some references discussed as Enterprise Architecture Roadmaps (EAM). Knowledge management is cross-cutting and has the same characteristics of corporate governance regardless of the technological framework to which it is applied.

These considerations of EA and MOT lead to a suggestion for a combined model where the business strategy is kept as a key motivator but suggesting a meta-strategy to indicate alternatives to formulated management strategies.

The suggested model is based on the EA3 framework (Bernard 2012). Static EA will only have to interact with business strategy and knowledge management. Given any initiation of change from business strategy or internal or external factors, an architecture management and transition plan must be used. Instead of a mapping based on strategy, business, and technology, the process is modified to initiate a broader innovation management process that needs to include issues of innovation portfolios, investments, competencies and qualifications, open or closed innovation, and other resources necessary (Brem & Voigt 2009; Brix 2014; Brown & Wyatt 2010). The innovation management process interacts with the technology intelligence process to ensure the highest level of information on available and potential solutions and conditions (Lee et al. 2013; Lichtenthaler 2004; Porter 2005). The innovation management process is followed by the technology roadmap which as a full or partial subset will include the EA roadmap (Gerdsri et al. 2009; Geum et al. 2011; Oliveira & Rozenfeld 2010). Knowledge management is cross-cutting along the timeline and is expected to interact with all processes. The management and transition plan must be expected to be exposed to iterations that may arise at any of the previous points: strategy, initiators, innovation management, technology intelligence, or even in misconceptions related to the current view of EA.

EA's Contribution to MOT

Apart from the contribution of MOT meta-disciplines to EA, it is worth considering the contribution and potential contributions from EA to MOT. Two issues are to be reviewed:

1. Traditionally, EA frameworks have indeed been IT-centric, and, therefore, in order to contribute to MOT, a more generic technology view must be employed.
2. Future business and technology models exhibit a dominance of integrated technological platforms with network-based capabilities.

For existing IT-centric EA models, these have limitations on the degree of abstraction from IT to, for example, mechanical technologies. The IT portion of generic technologies can obviously be included in EA frameworks. This does not mean that the EA3 framework (Bernard 2012) rules out the fact that technologies can be other technologies than IT; e.g., manufacturing technologies, transportation technologies, and communication technologies. For future technologies having generic technologies and IT deeply integrated, EA must be expected to be able to create meaningful governance frameworks. Reviewing the case of DigiCorp, although it is a digital services provider, it cannot utilize EA to its fullest due to the misalignment of applications, services, and data. The change from distributed master data to a central repository will enable all services and technologies to be encompassed by EA. The case of PayDK is likewise obvious in its shift from mixed analog and digital channels of citizen services to a single digital channel; EA will then be applicable on the transferred services. With Kapsali's (2011) suggestion of systems thinking as a significant factor to improve innovation projects, it is straightforward to assume that an integrated EA/MOT approach is an advantageous combination with more search for a holistic view of actions and priorities.

Selecting Model Perspectives

Having enterprise architects focus more broadly on the perspective of business and (information) technology than today may seem unachievable. However, when future business models combine "traditional" technologies with Internet-connected technologies, the enterprise architects must embrace broader and, presumably, include new competencies. In areas like mechanical engineering, it is expected that cars as well as any type of large engine have substantial processing power, and the vendor will sell the vehicles with a broad range of predictive fault monitoring features and may even have a business model delivering performance

instead of fixed assets. The enterprise architect must manage this.

CONCLUSION

The presented model suggests theoretical elements that would coordinate, not to say align, EA better with technological governance frameworks of the company.

As suggestions for further work, it is interesting to examine if the four meta-disciplines are sufficient and adequate. Idea management, Governance, Risk, and Compliance (GRC), systems of innovation, networks of innovation, intellectual capital, and communities of practice are potential meta-disciplines for further research. The consistent claim in EA as well as in the meta-disciplines to connect to business strategy is interesting but also questionable. Business strategies are typically more fluctuant than the actual business infrastructure and the legacy of systems and processes. Interesting for further research is thus also a more tactical, persistent, and/or “conservative” foundation for determining structures.

The discussions of this article are based on theory and the idea of solidifying EA by improving the theoretical underpinning. The discussions do have reservations. This research is qualitative and not verified by empirical sampling; e.g., EA practices are more MOT-oriented *versus* “traditional”. The argument that EA by and large is closely associated with MOT proves correct in the case companies, but it has not been tested on a greater number of companies; e.g., companies focusing on physical artefacts, which could be interesting.

The current article has demonstrated an extension of EA with four MOT meta-disciplines. The extension offers a broader interface between EA and other parallel activities of the enterprise. In future businesses, a combination of architecture and innovation is critical, and “end-to-end” architectural perspectives of networked product and services are attractive.

ABOUT THE AUTHOR

Torben Tambo was awarded an MSc in Computer Science from Danish Technical University in 1989 and has worked for 17 years in IT management roles. Torben serves as program coordinator for MSc in technology-based business development at Aarhus University. Research topics are within issues of strategic IT, IT management, software development, and business-technology issues.

REFERENCES

J. Afshani, A. Harounabadi, M. Dezfouli: A New Model for Designing Uncertain Enterprise Architecture, *Management Science Letters*, Vol. 2, No. 2, pp.689-696 (2012).

H. Arman, J. Foden: Combining Methods in the Technology Intelligence Process: Application in an Aerospace Manufacturing Firm, *R&D Management*, Vol. 40, No. 2, pp.181-194, (2010).

N. Arshed, J. Finch, R. Bunduchi: Technology Roadmapping and SMEs: A Literature Review, in: *DRUID Society Conference* (2012).

M.K. Badawy: *Developing Managerial Skills in Engineers and Scientists: Succeeding as a Technical Manager*, John Wiley & Sons (1995).

M.K. Badawy: A Research Architecture for Technology Management Education, *Engineering Management Review*, IEEE, Vol. 38, No. 3, pp.95-115 (2010).

C. Battistella: *The Organisation of Corporate Foresight: A Multiple Case Study in the Telecommunication Industry*, *Technological Forecasting & Social Change* (2013).

N.A. Behkami, T. Daim: Research Forecasting for Health Information Technology (HIT), *Using Technology Intelligence*, *Technological Forecasting & Social Change*, Vol. 79, No. 3, pp.498-508 (2012).

A. Bergek, S. Jacobsson, B. Carlsson, S. Lindmark, A. Rickne: Analyzing the Functional Dynamics of Technological Innovation Systems: A Scheme of Analysis, *Research Policy*, Vol. 37, No. 3, pp.407-429 (2008).

S. Bernard: *An Introduction to Enterprise Architecture*, AuthorHouse (2012).

P. Bernus: Enterprise Models for Enterprise Architecture and ISO 9000:2000, *Annual Reviews in Control*, Vol. 27, No. 2, pp.211-220 (2003).

H. Bouwman, H. van Houtum, M. Janssen, G. Versteeg: Business Architectures in the Public Sector: Experiences from Practice, *Communications of the Association for Information Systems*, Vol. 29, No. 1, pp.411-426 (2011).

C. Braun, R. Winter: A Comprehensive Enterprise Architecture Metamodel and its Implementation Using a Metamodeling Platform, *Desel, Jörg; Frank, Ulrich*, pp.24-25 (2005).

A. Brem, K. Voigt: Integration of Market Pull and Technology Push in the Corporate Front End and Innovation Management – Insights from the German Software Industry, *Technovation*, Vol. 29, No. 5, pp.351-367 (2009).

M.S. Brenner: Technology Intelligence and Technology Scouting, *Competitive Intelligence Review*, Vol. 7, No. 3, pp.20-27 (1996).

J. Brix: Exploring a Radical Innovation Project as Source of Change in Organizational Routines, *Organization Science*, Special Issue on Routine Dynamics (2014).

T. Brown, J. Wyatt: Design Thinking for Social Innovation, *Stanford Social Innovation Review*, Winter 2010, pp.31-35 (2010).

- A. Bryman, E. Bell: Business Research Methods, 3rd Edition, Oxford University Press, Oxford (2011).
- M. Carvalho, A. Fleury, A.P. Lopes: An Overview of the Literature on Technology Roadmapping (TRM): Contributions and Trends, *Technological Forecasting & Social Change*, Vol. 80, No. 7, pp.1418-1437 (2013).
- O. Castellanos, L.M. Torres: Technology Intelligence: Methods and Capabilities for Generation of Knowledge and Decision making, in: *Technology Management for Global Economic Growth (PICMET)*, 2010 Proceedings of PICMET'10 (pp.1-9), IEEE (2010).
- C. Chandra, S. Kumar: Enterprise Architectural Framework for Supply-chain Integration, *Industrial Management & Data Systems*, Vol. 10, No. 16, pp.290-304 (2001).
- M.M. Crossan, M. Apaydin: A Multi-Dimensional Framework of Organizational Innovation: A Systematic Review of the Literature, *Journal of Management Studies*, Vol. 47, No. 6, pp.1154-1191 (2010).
- K.C. Desouza, C. Dombrowski, Y. Awazu, P. Baloh, S. Papagari, S. Jha, J.Y. Kim: Crafting Organizational Innovation Processes, *Innovation: Management, Policy, & Practice*, Vol. 11, No. 1, pp.6-33 (2009).
- M.J. Donate, J.D. Sánchez de Pablo: The Role of Knowledge-oriented Leadership in Knowledge Management Practices and Innovation, *Journal of Business Research* (2014).
- H. Etzkowitz, L. Leydesdorff: The Dynamics of Innovation: from National Systems and "Mode 2" to a Triple Helix of University-Industry-government Relations, *Research Policy*, Vol. 29, pp.109-123 (2000).
- R. Fischer, S. Aier, R. Winter: A Federated Approach to Enterprise Architecture Model Maintenance, *Enterprise Modelling and Information Systems Architectures*, Vol. 2, No. 2, pp.14-22 (2007).
- R. Garcia, R. Calantone: A Critical Look at Technological Innovation Typology and Innovativeness Terminology: A Literature Review, *Journal of Product Innovation Management*, Vol. 19, No. 2, pp.110-132 (2002).
- A. Garg, R. Kazman, H.M. Chen: Interface Descriptions for Enterprise Architecture, *Science of Computer Programming*, Vol. 61, No. 1, pp.4-15 (2006).
- N. Gerdtsri, D.F. Kocaoglu: Applying the Analytic Hierarchy Process (AHP) to Build a Strategic Framework for Technology Roadmapping, *Mathematical and Computer Modelling*, Vol. 46, No. 7, pp.1071-1080 (2007).
- N. Gerdtsri, R.S. Vatananan, S. Dansamasatid: Dealing with the Dynamics of Technology Roadmapping Implementation: A Case Study, *Technological Forecasting & Social Change*, Vol. 76, No. 1, pp.50-60 (2009).
- Y. Geum, S. Lee, D. Kang, Y. Park: Technology Roadmapping for Technology-based Product-Service Integration: A Case Study, *Journal of Engineering and Technology Management*, Vol. 28, No. 3, pp.128-146 (2011).
- A.H. Gold, A. Malhotra, A.H. Segars: Knowledge Management: An Organizational Capabilities Perspective, *Journal of Management Information Systems*, 18(1), 185-214 (2001).
- J.L. Hervás-Oliver, F. Sempere-Ripoll: Disentangling the Influence of Technological Process and Product Innovations, *Journal of Business Research* (2014).
- K. Hjort-Madsen: Architecting Government Understanding Enterprise Architecture Adoption in the Public Sector, PhD Thesis, Copenhagen, The IT University (2009).
- M. Hoegl, A. Schulze: How to Support Knowledge Creation in New Product Development: An investigation of Knowledge Management Methods, *European Management Journal*, Vol. 23, No. 3, pp.263-273 (2005).
- C. Holmes, M. Ferrill: The Application of Operation and Technology Roadmapping to Aid Singaporean SMEs Identify and Select Emerging Technologies, *Technological Forecasting & Social Change*, Vol. 72, No. 3, pp.349-357 (2005).
- H.-F. Hung, H.-P. Kao, Y.Y. Chu: An Empirical Study on Knowledge Integration, Technology Innovation, and Experimental Practice, *Expert Systems with Applications*, Vol. 35, pp.177-186 (2008).
- A.Ø. Jensen: Government Enterprise Architecture Adoption: A Systemic-Discursive Critique and Re-conceptualization, Copenhagen Business School (2010).
- J.A. Johannessen, B. Olsen, J. Olaisen: Organizing for Innovation, *Long Range Planning*, Vol. 30, No. 1, pp.96-109 (1997).
- F. Joubert: Knowledge Assets in Enterprise Architecture, Doctoral Dissertation, Stellenbosch, University of Stellenbosch (2010).
- M. Kahn: Archetypes of Organization: Laying Systemic Enterprise Architecture Foundations at an Upstream Oil and Gas Company, *Journal of Enterprise Architecture*, Vol. 8, No. 3, pp.45-61 (2012).
- D. Kang, J. Lee, K. Kim: Alignment of Business Enterprise Architectures using Fact-based Ontologies, *Expert Systems with Applications*, Vol. 37, No. 4, pp.3274-3283 (2010).
- M. Kapsali: Systems Thinking in Innovation Project Management: A Match that Works, *International Journal of Project Management*, Vol. 29, No. 4, pp.396-407 (2011).
- C.I.V. Kerr et. al.: A Conceptual Model for Technology Intelligence, *International Journal of Technology Intelligence and Planning*, Vol. 2, No. 1 (2006).

- C. Koch et al.: Institutionalized Technological Foresight – Methods, Impact, and Role in Policy-making and Business Development, in B. Abu Hijleh, M. Arif, T. Khalil, Y. Hosni (Eds): Proceedings of the 17th International Conference on Management of Technology (IAMOT 2008).
- C. Koch, M. Olsen, T. Tambo: How Little do we Need to Know about Enterprise Resource Planning (ERP)? – A Critical Review of Information Systems Research on ERP, Proceedings of the 33rd Information Systems Research Seminar in Scandinavia (2010).
- N. Kock (Ed): Information Systems Action Research – An Applied View on Emerging Concepts and Methods, Integrated Series on Information Systems, Vol. 13 (2007).
- J.J. Korhonen, J. Poutanen: Tripartite Approach to Enterprise Architecture, Journal of Enterprise Architecture, Vol. 9, No. 1, pp.28-38 (2013).
- M.M. Lankhorst: Enterprise Architecture Modeling – The Issue of Integration, Advanced Engineering Informatics, Vol. 18, No. 4, pp.205-216 (2004).
- J. Lapalme: Three Schools of Thought on Enterprise Architecture, IT Professional, 14(6), 37-43 (2012).
- V. Lazzarotti, R. Manzini: Different Modes of Open Innovation: A Theoretical Framework and an Empirical Study, International Journal of Innovation Management, 13(04), 615-636 (2009).
- C. Lee, B. Song, Y. Park: An Instrument for Scenario-based Technology Roadmapping: How to Assess the Impacts of Future Changes on Organizational Plans, Technological Forecasting & Social Change (2014).
- J.H. Lee, H.I. Kim, R. Phaal: An Analysis of Factors Improving Technology Roadmap Credibility: A Communications Theory Assessment of Roadmapping Processes, Technological Forecasting & Social Change, Vol. 79, No. 2, pp.263-280 (2012).
- S. Lee, M. Cho, S.K. Song, H. Jung: On Services and Insights of Technology Intelligence Systems, in Human Interface and the Management of Information, Information and Interaction for Learning, Culture, Collaboration, and Business, (pp.579-587), Springer Berlin Heidelberg (2013).
- E. Lichtenthaler: Technological Change and the Technology Intelligence Process: A Case Study, Journal of Engineering and Technology Management, Vol. 21, No. 4, pp.331-348 (2004).
- E. Lichtenthaler: Managing Technology Intelligence Processes in Situations of Radical Technological Change, Technological Forecasting & Social Change, Vol. 74, No. 8, pp.1109-1136 (2007).
- T. Magoulas, A. Hadzic, T. Saarikko, K. Pessi: Alignment in Enterprise Architecture: A Comparative Analysis of four Architectural Approaches, Electronic Journal Information Systems Evaluation, Vol. 15, No. 1 (2012).
- L. Mathiassen, P.A. Nielsen: Engaged Scholarship in IS Research, Scand., Journal of Information Systems, Vol. 20, No. 2 (2008).
- R.E. Miles, C.C. Snow: Organization Theory and Supply Chain Management: An Evolving Research Perspective, Journal of Operations Management, Vol. 25, No. 2, pp.459-463 (2007).
- C. Møller, C.J. Brandt, A. Carugati: Deliberately by Design, or?, Enterprise Architecture Transformation at Arla Foods, Advances in Enterprise Information Systems II, Vol. 91 (2012).
- J. Nill, R. Kemp: Evolutionary Approaches for Sustainable Innovation Policies: From Niche to Paradigm?, Research Policy, Vol. 38, No. 4, pp.668-680 (2009).
- I. Nonaka, H. Takeuchi: A Theory of Organizational Knowledge Creation, IJTM, Special Publication on Unlearning and Learning, Vol. 11, Nos. 7/8 (1996).
- O. Noran: An Analysis of the Zachman Framework for Enterprise Architecture from the GERAM Perspective, Annual Reviews in Control, Vol. 27, No. 2, pp.163-183 (2003).
- P. Närman, H. Holm, M. Ekstedt, N. Honeth: Using Enterprise Architecture Analysis and Interview Data to Estimate Service Response Time, The Journal of Strategic Information Systems, Vol. 22, No. 1, pp.70-85 (2013).
- OECD: Oslo Manual: Guidelines for Collecting and Interpreting Innovation Data, OECD publishing (2005).
- M.G. Oliveira, H. Rozenfeld: Integrating Technology Roadmapping and Portfolio Management at the Front-end of New Product Development, Technological Forecasting & Social Change, Vol. 77, No. 8, pp.1339-1354 (2010).
- M.K. Peter, D.G. Jarratt: The Practice of Foresight in Long-term Planning, Technological Forecasting & Social Change (2013).
- I.J. Petrick, A.E. Echols: Technology Roadmapping in Review: A Tool for Making Sustainable New Product Development Decisions, Technological Forecasting & Social Change, Vol. 71, No. 1, pp.81-100 (2004).
- A. Pettigrew, R. Woodman, K. Cameron: Studying Organizational Change and Development – Challenges for Future Research, Academy of Management Journal, Vol. 44, pp.697-713 (2001).
- R. Phaal, C.J.P. Farrukh, D.R. Probert: Technology Management Tools: Concept, Development, and Application, Technovation, Vol. 26, No. 3, pp.336-344 (2006).
- R. Phaal, C.J.P. Farrukh, D.R. Probert: T-Plan – The Fast-Start to Technology Roadmapping: Planning your Route to Success, Institute for Manufacturing, University of Cambridge, Cambridge, UK (2001).
- R. Phaal, G. Müller: An Architectural Framework for Roadmapping: Towards Visual Strategy, Technological Forecasting & Social Change, Vol. 76, pp.39-49 (2009).

- R. Phaal, C.J. Farrukh, D.R. Probert: Technology Roadmapping – A Planning Framework for Evolution and Revolution, *Technological Forecasting & Social Change*, Vol. 71, No. 1, pp.5-26 (2004).
- S.A. Phillips: Trade Association Strategies for Providing Technology Intelligence to Small and Medium sized Enterprises – A Study of UK Technology Foresight Processes, Brunel University (2010).
- A.L. Porter: QTIP: Quick Technology Intelligence Processes, *Technological Forecasting & Social Change*, Vol. 72, No. 9, pp.1070-1081 (2005).
- M. Pulkkinen: Systemic Management of Architectural Decisions in Enterprise Architecture Planning: Four Dimensions and Three Abstraction Levels, in *System Sciences*, 2006, HICSS'06, Proceedings of the 39th Annual Hawaii International Conference on (Vol. 8, pp.179a-179a) IEEE (2006).
- J.W. Ross: Creating a Strategic IT Architecture Competency: Learning in Stages, MIT Sloan Working Paper No. 4314-03, MIT Sloan School of Management (2003).
- J.W. Ross et al.: Enterprise Architecture as Strategy: Creating a Foundation for Business Execution, Harvard Business School Press (Chapters 2 and 3) (2006).
- M. Schilling: Strategic Management of Technological Innovation, 4th Edition, McGraw-Hill (2012).
- M. Schultz, M.J. Hatch: Living with Multiple Paradigms: The Case of Paradigm Interplay in Organizational Culture Studies, *The Academy of Management Review*. Vol. 21, pp.529-55 (1996).
- J. Sundbo, F. Gallouj: Innovation as a Loosely-coupled System in Services, *International Journal of Services Technology and Management*, Vol. 1, No. 1, pp.15-36 (2000).
- H. Takeuchi, I. Nonaka: The New New Product Development Game, *Harvard Business Review* (1986).
- M. Taleb, O. Cherkaoui: Pattern-oriented Approach for Enterprise Architecture: TOGAF® Framework (2012).
- T. Tamm et al.: Delivering Business Value Through Enterprise Architecture, *Journal of Enterprise Architecture*, Vol. 7, No. 2, pp.17-30 (2011).
- A.H. Van de Ven: Central Problems in the Management of Innovation, *Management Science*, Vol. 32, No. 5, pp.590-607 (1986).
- B. Van Looy, T. Martens, K. Debackere: Organizing for Continuous Innovation: On the Sustainability of Ambidextrous Organizations, *Creativity and Innovation Management*, Vol. 14, No. 3, pp.208-221 (2005).
- J. Vanston: Better Forecasts, Better pPlans, Better Results, *Research Technology Management*, pp.47-58 (Jan-Feb 2003).
- M. Veugeliers, J. Bury, S. Viaene: Linking Technology Intelligence to Open Innovationb Technological Forecasting & Social Change, Vol. 77, No. 2, pp.335-343 (2010).
- G. von Krogh, N. Geilinger: Knowledge Creation in the Eco-system: Research Imperatives, *European Management Journal*, Vol. 32, No. 1, pp.155-163 (2014).
- G. Walsham: Interpretive Case Studies in IS Research: Nature and Method, *European Journal of Information Systems*, Vol. 4, pp.74-81 (1998).
- J.A. Zachman: A Framework for Information Systems Architecture, *IBM Systems Journal*, Vol. 26, No. 3, pp.276-292 (1987).

Article

From Enterprise Architect to Opportunity Architect: The Changing Role of Enterprise Architecture in a Digital Transformation Context

Greet Bontinck, Prof. Dr. Bjorn Cumps, Prof. Dr. Stijn Viaene, Dr. Wesley Bille, and Joachim Vanden Brande

Abstract

A digital transformation is felt in every fibre of the organization. In order to deal with the challenges that come with such a transformation initiative, one-off point solutions are not enough, but a more systemic, architecture-driven approach is needed. What does digital transformation mean for the enterprise architect? Through a multiple case study approach, this research aims to gain insights into the changing role of enterprise architects in a digital transformation context, as well as to identify the new challenges and opportunities arising in this regard. Today, enterprise architects are at a crossroads: the digital transformation projects in their organizations have rendered them more valuable. However, the key question is whether they will focus on enabling and support, or whether they will move one step beyond, leading the way, becoming true opportunity architects.

Keywords

Enterprise Architecture, Digital Transformation, Case Study

THE CHANGING ROLE OF EA IN A DIGITAL TRANSFORMATION CONTEXT

In times where digitization is radically transforming many industries, incumbent organizations are trying to grasp what the digital age means for their current business model. In order to stand out, the development of the following digital capabilities is key (Bossert et al. 2014):

- Developing digital product innovation skills in order to meet changing customer expectations in a digital world
- Providing a seamless (digital and physical) multi-channel experience
- Using big data and advanced analytics to better understand customer behavior
- Improving the automation of operations and the digitization of business processes

While, until a few years ago, some organizations believed that adding a few nicely designed front-faced tools, or a focus on process automation, would be sufficient to keep abreast of the digital revolution, organizations are now moving beyond this limited digital focus. Today, we see that digital transformation is approached from:

1. The customer perspective
2. The employee perspective
3. A collaboration perspective

Moreover, organizations face speed-to-market pressures, forcing them to quickly adapt to the new

digital environment. In order to be able to develop these capabilities and to respond to the challenges of the digital world, organizations must have the right processes, structures, IT systems, roles, and incentives in place (Ross et al. 2014). As these elements all interact with one another, this is where Enterprise Architecture (EA) as a management discipline comes into play.

As the discipline of EA is maturing, the role of enterprise architects has now moved beyond IT architecture:

"IT architecture was a useful introduction to the practicing of EA within companies, but that was just the start of it." (Ross et al. 2014)

EA is now helping to manage an organization's complexity, and enabling an understanding of the structure and behavior of the organization's business and IT from a holistic viewpoint. Moreover, the discipline of EA can help the organization's decision-makers creating the roadmap from strategic goals through project investments to measurable business value and performance improvements (Bente et al. 2012).

It is expected that in the coming years organizations capable of developing strong EA competencies may develop an important competitive advantage in the digital economy (Ross et al. 2014). Thus, the current role of enterprise architects in a digital transformation context warrants further research.

“TO TRANSFORM OR NOT TO TRANSFORM” IS NOT THE QUESTION

Research Design

During October and November 2015, we conducted 13 case studies on the role of enterprise architects in digital transformation. This research was jointly conducted by the Vlerick Centre for Excellence in Enterprise Architecture and our partner LoQutus.

The organizations selected for the case studies were characterized by:

1. A minimum level of EA maturity
2. An ongoing “digital transformation”

“Digital transformation” was defined as:

“... a specialized type of business transformation where IT plays a dominant role, and where an organization’s strategy, structure, culture, and processes are transformed by using the potential and power of digital media and the Internet.” (Uhl et al. 2014)

The sample selection covers various sectors (Banking, Payments & Insurance, Transport, Energy, Retail & Distribution, Government, and Telecom), and its overall EA maturity varies between “Foundation” (the organization has an enterprise-wide view and a plan for technology, which are used to align the organization’s business and technology) and “Extended” (the science, tools, and techniques of EA are extended into (and used by) all parts of the enterprise to design/describe much more than technology) (Gøtze 2013).

In total, 17 semi-structured interviews were conducted with employees in charge or closely involved in their organization’s digital transformation, each with a clear view on the role of EA in this context. The final interviewee selection included roles such as CIO, E-transformation Director, Enterprise Architect, Head of Architecture, Innovation Manager, and Director of IT. Thus, both the business side as well as the enterprise architect practitioners’ side was represented in the interviews.

Overview case studies	
13 case studies	Acerta, Agentschap voor Onderwijsdiensten, Axa, CM, Colruyt, Eandis, ING, KBC, MasterCard, NMBS, P&V Group, Proximus and VDAB
17 semi-structured interviews	Employees in charge or closely involved in the digital transformation and with clear view on the role of EA
Covered sectors	Banking, Payments & Insurance, Transport, Energy, Retail & Distribution, Government, Telecom
Case study selection criteria	• Minimum level of EA maturity • Involved in enterprise-wide digital transformation

Figure 1: Overview Case Studies

Interviewees were questioned about the main challenges their organizations are currently facing in a fast-moving digital world, the characteristics of the digital transformation of their organization, the parties initiating and/or involved in the digital transformation project, and, most importantly, the role of enterprise architects in this context. Our analysis provides valuable insights into the extent to which enterprise architects are involved in digital transformation, including the scope of their tasks and responsibilities, and the impact of EA on an organization’s ongoing transformation.

The Perspective on Digital Transformation is Maturing

All interviewed organizations reported that they currently face challenges which force them to revise their current way of operating. These challenges come in different forms: becoming a “paperless” organization, moving beyond commodity services and creating value propositions for a digital world, introducing digital customer touchpoints, adapting digital innovations to regulatory requirements, or transforming the existing infrastructure in line with new technologies.

The nature of the observed digital transformations can be grouped into several digital transformation types (Figure 2), whereby we see a strong focus on the customer perspective (e.g., digital customer touchpoints), the employee perspective (e.g., becoming a paperless organization), and the collaboration perspective (e.g., open data). The size of each circle gives an indication of the frequency of each observed digital transformation type. In several case studies, more than one digital transformation type was identified. It is clear that the main focus of digital transformation projects focuses on customer interaction: creating digital touchpoints for customers, capturing data, and leveraging it for better targeting. Digital transformation seems dominated by projects from the marketing realm.

As different as the challenges of each organization may be, none of the interviewees questioned the need to digitally transform. All interviewed organizations reported they were involved in digital transformation initiatives with a profound impact on the organization. In our previous study we reported the end of corporate-wide, top-down large EA projects. EA on a project-by-project basis gained momentum. Today, EA is again, either in a supporting or leading role, part of large, corporate-wide digital transformation programs.

The case studies show that organizations are coping with the digital transformation challenge by addressing the organization’s needs at a systemic level. Using this approach, the entire set of interacting components of the enterprise, such as the organization’s strategy, organizational structure, processes, applications, and

data and technology stack are taken into account. Enterprise architects are consulted when connecting front and back, developing open APIs, setting-up data lakes, and building the new infrastructure for the organizations. Great stuff, but again, mainly IT-focussed. Yet, we did come across opportunities for enterprise architects to move beyond their traditional IT realm.

The next section assesses the role of EA in a digital transformation context, as the above-described holistic approach may pose an opportunity for enterprise architects to collaborate actively and guide the way. Moreover, multiple risks and opportunities for EA practitioners in this context are identified.

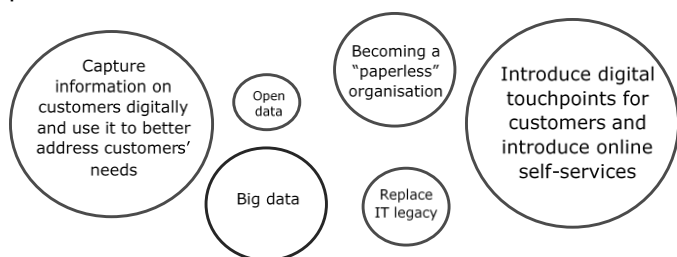


Figure 2: Categorization of Digital Transformation Types

THE ROLE OF EA IN A DIGITAL TRANSFORMATION CONTEXT: FINDINGS, OPPORTUNITIES, AND CHALLENGES

The analysis of the interview data resulted in six main findings on the role of EA in a digital transformation context. Further, several related challenges and opportunities were identified for EA practitioners in this context. The analysis was concluded with a validation of the results during a feedback session of the Centre of Excellence in EA at Vlerick Business School, attended by 12 EA practitioners from various industries.

Figures 3-6 and 8-9 summarize the main research findings, as well as one opportunity and one risk related to each specific finding. It must be emphasized that the list of findings and related challenges and opportunities is by no means exhaustive. However, this section aims to provide an overview of the most striking conclusions with regard to EA in a digital transformation context, which were consistently observed in the case studies and further validated.

Finding 1: Digital Transformation is Becoming a Priority of the Executive Committee

As digital transformation initiatives are becoming more strategic and enterprise-wide, they are being closely monitored by the organization's executives. The parties leading or involved in the digital transformation projects were often selected by the executive committee, to whom they directly report. This evolution paves the way

for EA to gain visibility by the organization's executive committee, and to prove its added value in a digital transformation context.



Figure 3: Finding 1 and Related Opportunity and Challenge

The case studies demonstrate that the view of executives on EA and its added value is crucial as to whether enterprise architects are formally involved in the organization's digital transformation projects. A clear example can be found in one of the case studies, where an enterprise architect commented:

"The executive committee acknowledged that the existing silo structure was a major obstacle in realizing the intended customer-centricity, and more generally, our organization's digital transformation. Here, EA got in the picture as a way to adapt the organization's silo-structure."

Using this "foot in the door" of the intended digital transformation and taking on a proactive attitude may provide an opportunity for increased EA involvement in the transformation programs as of the initial phase. As digital transformation is high on the executive committee's agenda, it is up to EA to put itself on the map here as a management discipline capable of guiding the organizational, process, and technology changes that come with a digital transformation effort at the enterprise-wide level. It is a challenge for enterprise architects not to "miss the digital boat" and take on a proactive approach in ensuring EA involvement in their organization's digital transformation.

Finding 2: Enterprise Architects are Getting Access to the Board Room

A previous study, conducted in 2013, reported that C-level executives generally perceived EA as being too technical, formal, and prescriptive, relying on complicated techniques and elaborate frameworks. Moreover, due to its origins in the IT domain, the

business value of EA had not been widely established (Cumps et al. 2013).

Today, we see that this perception has evolved in several organizations. 8 out of 13 organizations reported that EA is finding its way to the board room: in one of the case studies, the organization's CEO even declared that re-designing the organization's architecture was a top priority, and that investments had to be made in designing the right architecture in order to successfully implement the ongoing digital transformation. More generally, the analysis of the interview data showed a clear trend that C-level executives have now become more acquainted with EA as a management discipline. One e-transformation director illustrated this trend as follows:

"Our enterprise architects are increasingly invited by the executive committee for advice and assistance when defining the organization's digital strategy and translating this strategy to a future-proof organizational architecture."

When asked about the "keys to success" in this matter, the following elements were mentioned consistently:

- Having a (chief) enterprise architect in the organization who is able to translate EA concepts to the board's language is invaluable in order to convey the added value of EA in a digital transformation context. This requires strong communications skills, and the ability to build up a network within the organization. A strong business partner can take up a similar role to connect and promote EA with the organization's senior management.
- The EA discipline within the organization has now reached a sufficient level of maturity, providing EA with a proven track-record of adding value. In several organizations where senior management is now a loyal supporter of EA, it was argued that this support emerged after an initial successful intervention of architects in various projects. One Head of EA underlines that *"by taking on an enabling role, EA has been able to demonstrate its potential to add substantial value in comparison to the previously adopted transformational approach where no models or structure supported the change. Thus, architects are able to build credibility and demonstrate how EA can become the board's partner in guiding structural changes within the organization"*. Such an approach can contribute to raising awareness within the board that "gut-feel" does not suffice to successfully complete an organization's digital transformation, and that EA has value to add here.
- All organizations where EA is on the executive committee's agenda reported that important success factors in this regard are personal relations and beliefs: the former chief architect who now

directly reports to the board, or informal contacts which are well-established between an enterprise architect and a board member.

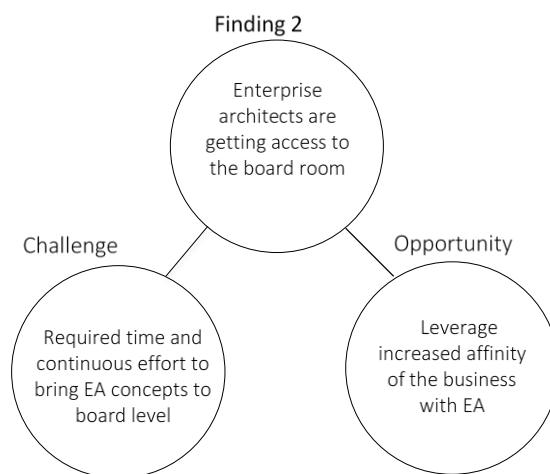


Figure 4: Finding 2 and Related Opportunity and Challenge

Several case studies showed that senior management support is key in order to convey the added value of EA to the business-side of the organization. In a top-down approach, where the executive committee or CEO obliges project managers to sit around the table with enterprise architects before the start of a transformational project, it was reported that this obligation creates an opportunity for the business to get acquainted with EA and, at best, its benefits. This obligatory involvement of architects creates affinity with EA within the organization, which can be leveraged. In organizations where EA has not found its way to the board (yet), the roles assigned to enterprise architects in the digital transformation context were consistently reported to be more limited.

One interviewed innovation manager described this effect as follows:

"At first, the only incentive for a project manager to invite enterprise architects to the table and discuss my projects was to follow the procedure imposed by our CEO and obtain the project approval. Now, our incentive to do so is that we have experienced in the past that the involvement of enterprise architects prevents us to make stupid decisions."

In multiple case studies, it was highlighted that one of the main challenges faced in order to bring EA to board level was the investment of time and continuous effort required to adjust EA concepts to the board members' knowledge of the discipline, and to build a good relationship between the EA team and the board. Here, it is important that this role be taken up by the people within the EA team who possess the right communication skills.

Finding 3: When Architects are Invited to the Digital Party, they Often Leave Early

The case studies revealed that the involvement of enterprise architects in large transformational programs is generally established. In 12 out of 13 case studies, architects were reported to be involved – to a greater or lesser extent – in their organization’s digital transformation efforts. However, the interview data shows that, in a majority of the case studies, an enterprise architect’s tasks and responsibilities are mostly situated at the start of the project, and mainly include scope definition, budgeting decisions and requirements setting.

When asked about the architect’s role during the next phases of the transformation, most interviewees agreed that this role is mainly limited to doing a minimal follow-up of the projects. A common argument put forward was that an architect’s role in this phase can be characterized as the “auditor”, where the architect follows the projects from a distance, acts as a gatekeeper, and guards whether the requirements and scope set at the start of the project are complied with.

Nevertheless, several interviewees indicated that this phase is where the architect can make a difference: in setting things right from the start. In organizations where the involvement of architects at the later stages of the digital transformation is not limited to a gatekeeping function, the interviewed architects report that playing a more extensive role in the implementation teams is highly relevant. The role of architects as a full member of the implementation teams is profoundly different, as they share the team’s responsibility to deliver the digital transformation. The Head of EA in one of the interviewed organizations, clearly convinced of the added value of this approach, stated:

“Close collaboration with the business during the implementation phase allows the architects to collect information on how the predefined architectural requirements are realized in practice, and to see what works and what does not.”

This approach allows architects to challenge the implementation team on the impact of its decisions on the organization’s architecture. Architects can take this real-world evidence to the next architectural exercise, which allows them to closely align the organization’s blueprint architecture with the business reality.

The involvement of architects at the start of digital transformation projects provides the opportunity for architects to extend their involvement into the implementation phase. Here, it is key for architects to not limit their involvement to simply ticking the boxes, and to act as a controller. Instead, the architect can contribute by taking on a closed-learning approach and

becoming a coach during the implementation phase of the digital transformation projects. Moreover, this provides the opportunity to spread EA throughout the organization. Enterprise architects should keep their eyes open for opportunities in this regard, and take on a proactive role.

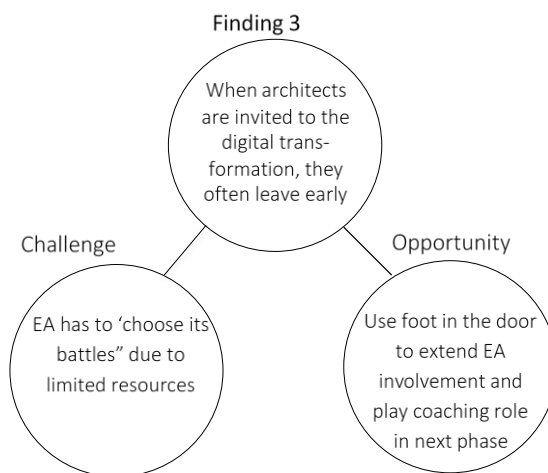


Figure 5: Finding 3 and Related Challenge and Opportunity

Such a role is not yet a reality in most organizations. One enterprise architect commented during the feedback session on the research that “EA has to choose its battles”, not only due to the limited resources of the EA team, but also due to the fact that architects often do not feel comfortable or possess the right skills to play a more extended role throughout the transformation. Another challenge put forward in this regard, is that several architects reported to be comfortable in their role in the preliminary study phase of the project. As such, many architects are bound to leave the digital party early.

Finding 4: Enterprise Architects are Gradually Accepting Semi-formal Methods

“Visualizing an architectural framework may look impressive, but it is worth nothing if it is not applied later on” says one CIO, commenting on the finding that architectural models and frameworks are generally perceived by the business as being overly complex and technical. As the business feels pressure to accelerate the organization’s digital transformation, interviewees from the business side report that they perceive the use of such frameworks as a delaying factor. Also, these frameworks and models are often not understood by the business. Thus, they should be avoided. In line with the trend of lightweight governance, several EA teams reported they had adopted a semi-formal approach to EA, where the enterprise-wide architectural models and frameworks are used less often and where architects

translate specific conclusions into easy-to-understand models in a common language. The tagline here can be summarized as: “Semi-formal is good enough”.

When asked about the opportunity of using EA methods in a less rigid way, interviewees consistently responded that it creates buy-in from the business. By re-using existing terminology used by the business, several interviewees reported that this approach leads to a further spread of the use of EA throughout the entire organization. Instead of pushing EA models and frameworks top-down, architects can use this approach to architect the organization in collaboration with the business. A clear example of this approach can be found in one of the case studies, where the enterprise architect stated that:

“Before, formal EA-models and frameworks were forced upon the business in a top-down way, and architects could design EA models independently. Now, they are adopting a more collaborative approach, where architects must discuss the organization’s architecture with the business, and must explain the designs to the business. Nevertheless, this can be a frustrating process.”

These findings build further on the research results of the 2013 study, where it was concluded that organizations no longer believed in large-scale top-down EA programs, and where a more pragmatic EA approach was advised, in which the enterprise-wide picture is still outlined at a high-level, but where EA focuses on deliverables that are useful and used in the organization (Cumps et al. 2013).

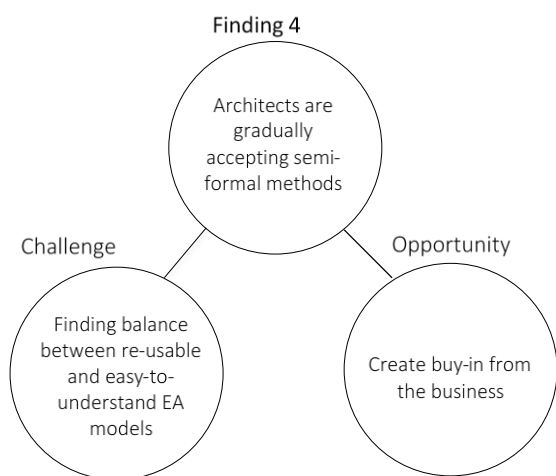


Figure 6: Finding 4 and Related Challenge and Opportunity

The analysis of interview data shows that this evolution brings new challenges. In organizations where the EA team has been adopting such an EA approach, the architects reported that they are struggling with finding their role in this new – less formal – approach, and with abandoning the models and frameworks that were used before. Also, a drawback to this approach is that the EA

models are often created *ad hoc*, and are not actively maintained and consolidated. This makes it difficult to re-use them or integrate them in the organization’s architectural models. Architects must find a balance between creating useful EA models, and models which are easy-to-understand, but very unstructured and difficult to re-use or integrate in the enterprise architecture later on. A guiding principle here is the ultimate goal is not to design the perfect architecture, but the most future-proof architecture, adapted to the organization’s business reality and strategy.

Finding 5: Architects are Struggling with Innovation

“Innovation” is the process of turning opportunities into new, value-generating ideas, and positioning those ideas into widely used practices (Tidd & Bessant 2009). This process can have different degrees of impact on the organization, varying from incremental innovation to radical innovation (Safrudin et al. 2014). The case studies revealed that enterprise architects play a role in the incremental innovation process (for example, an update of IT applications), while this is less the case for more radical innovations. Architects are rather perceived by the business as enablers of innovation than creators of the latter type of “innovation”, which are often front-faced and marketing-driven. The interviewed enterprise architects stated that, when it comes to the set-up of labs and agile experimentation, they are struggling to position themselves.

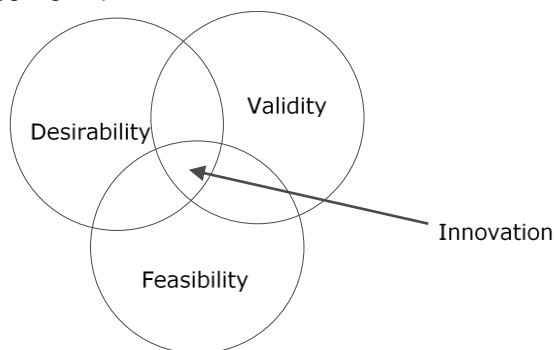


Figure 7: Dimensions of Business Opportunities Assessment

In the process of innovation and exploration of new business opportunities, potential products, and services are to be found in the intersection of desirability (“Does the proposed product/service fill a customer need, and do our customers actually want the product/service?”), feasibility (“Does our organization have the required capabilities of delivering this product/service?”), and validity (“How does this product/service deliver value to the business?”), as indicated in Figure 7.

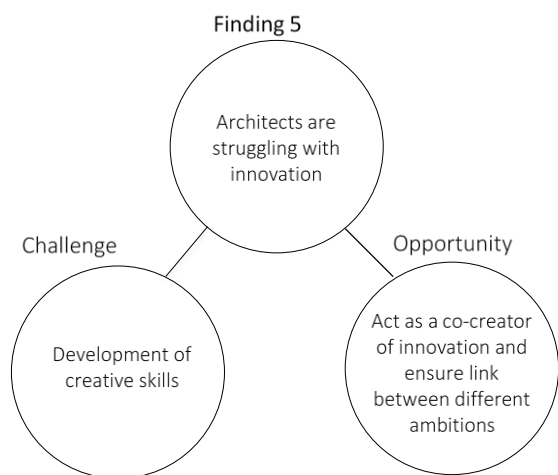


Figure 8: Finding 5 and Related Challenge and Opportunity

In our 2013 study, we concluded that the role of enterprise architects in this regard was typically located in the “feasibility” segment (Cumps et al. 2013). While today, architects are still mainly active in the feasibility segment, the analysis of the interview data demonstrates that the role of enterprise architects in innovation is now also entering the validity segment. In some cases, the role of architects in innovation was reported to be non-existent, whereas in the majority of organizations, the role of architects was mainly doing feasibility studies (the feasibility dimension) and assessing whether new ideas of the business fit the overall strategy of the organization (the validity dimension). When interviewees were asked whether architects would be able to contribute to the desirability segment, this was not seen as a strength of architects.

Nevertheless, several interviewed architects have expressed their ambition to contribute to their organization’s (more radical) innovation processes in a more substantial way than they are currently doing. In this regard, we see opportunities for enterprise architects to add value in:

1: The articulation of a clear innovation ambition

An organization’s endogenous innovation ambition is aimed at activating the hidden growth potential of the organization’s existing products or services (Scheer & Burns 2015). This innovation ambition can either take the form of (1) optimizing existing products for the existing customers, or (2) the introduction of “adjacent innovations”, leveraging existing capabilities in other domains (Nagji & Tuff 2012).

When playing on the level of optimization, the enterprise architect’s involvement adds value with regard to operational excellence, by operating in a “constraint-mode”, which means that his role is understanding the impact of the proposed changes on the organization’s

different departments, activities, and processes. However, if the organization’s innovation ambition relates to adjacent innovations, the enterprise architect’s oversight and insight in the organization make him well-placed to map the existing capabilities in a structured way, as well as to provide direction in redesigning the existing organizational structures. More general, the enterprise architect can add value by operating in an “option-mode”, by providing guidance as to which adjacent innovations on the table are feasible to implement, given the organization’s existing asset base.

The exogenous innovation ambition, on the other hand, is focused on identifying external assets offering opportunities for innovation (Scheer & Burns 2015). This innovation ambition typically aims at the development of “breakthrough products”, where the benchmark used is not the existing organization, but other disrupting companies. On this level of the innovation ambition, the enterprise architect adds value by operating in an “opportunity-mode” and by thinking out-of-the-box.

2: Ensuring the link between different innovation ambitions

Enterprise architect involvement is not only valuable with regard to realizing the above-mentioned innovation ambitions, but also with regard to ensuring the link between their organization’s different innovation ambitions.

Examples of an architect’s contribution in this regard are the identification of assets that have previously been developed in order to strengthen the core which can be leveraged for disruptive products; or the development of a strategy to push disruptive products that were developed in a lab into the core organization.

One interviewee commented:

“In order to ensure the link between all ongoing and newly proposed innovation initiatives within our organization, we are planning the set-up of an “Architecture Board”.”

3: Participating in strategic discussions

Given the fact that enterprise architects have excellent oversight and insight at enterprise level, there is an opportunity for architects to act as a partner of the business in shaping the vision of in what direction the organization must evolve, while connecting the different departments, units, and silos within the organization. Together with the business, they are well-placed to participate in several important strategic discussions, such as the feasibility of scaling the innovation projects, or which investments should be added to the investment portfolio.

When contributing to innovation, creative skills are essential. However, surprisingly, when questioning chief architects about essential skills they look for when hiring

an architect, “creative skills” were not often mentioned. Therefore, it was identified to be a challenge to include creative profiles in EA teams, who also possess excellent technical and analytical skills.

Finding 6: Bimodal IT is a Major Challenge for Architects

The subject of coping with bi-modal IT proved to be unavoidable, as such an IT landscape was omnipresent in the case studies. Gartner defines “bimodal IT” as the practice of managing two separate coherent modes of IT delivery, one focused on stability and the other on agility.¹ Mode 1 is traditional and sequential, emphasizing safety and accuracy. Mode 2 is exploratory and non-linear, emphasizing agility and speed.

Overall, architects reported to be well acquainted with mode 1, while this is far less the case for mode 2, which often takes the form of front-facing systems and technologies emerging next to the legacy systems of mode 1. In the majority of the case studies, enterprise architects are not involved at all during the set-up of these front-end innovations. In a few other organizations, efforts are made to introduce an agile way of working in the environment of traditional IT systems with EA involvement, but this is not running smooth yet. However, in the context of a digital transformation, it will be key to connect both systems, as the exchange of information between both is one of the prerequisites to achieve customer-centricity. In order to achieve outside-in thinking and great customer experiences, a solid and future-proof architecture is needed for both the front-end and back-end of the organization.

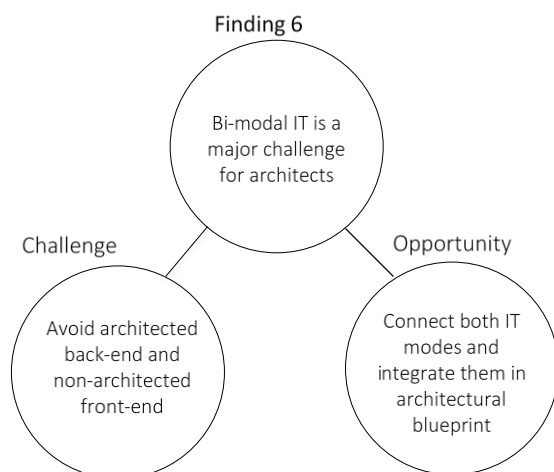


Figure 9: Finding 6 and Related Challenge and Opportunity

This context provides an opportunity for enterprise architects, who understand the complexity of the existing

legacy systems, to connect both systems and to integrate them in a future-proof architectural blueprint. Architects must avoid postponing the integration problem, and must take the initiative to get a grip on the bi-modal IT structure.

The challenge relating to bi-modal IT is to avoid an IT structure with an architected back-end and a non-architected front-end. It was marked as a major future challenge to integrate these front-faced IT systems with the legacy systems. One architect referred to these mode 2 systems as “shadow IT”, as they had been developed by the business side of his organization, who were determined to move forward fast and in an agile way, without involving the IT department or its enterprise architects, who were perceived by them as being more rigid and slow-paced.

FROM ENTERPRISE ARCHITECT TO OPPORTUNITY ARCHITECT

Enterprise architects are at a crossroads. Digital transformation is high on many executive committees’ agendas. Increased customer-centricity, experience- and data-driven business, new tech-savvy competitors, a faster pace of innovation, experimentation, agile/design-driven ways of working, and increased importance of ecosystems all blend together in a perfect storm called digital transformation. Organizations cannot tackle these challenges anymore with one-off point solutions. Clearly, a more systemic, architecture-driven approach is needed. Our research shows that many organizations are starting to feel this in every fibre of the company. As such, the stage is set. It is up to the enterprise architect to decide whether they will be in the play or not. And, more importantly, which role they will play. Based on our interview data, we see an emerging need for a new type of enterprise architect for digital transformation: the opportunity architect.

How does an opportunity architect differ from a traditional enterprise architect? We identified the following characteristics in our cases.

They coach the business.

It is clear from the cases that organizations tend to overestimate the short-term and underestimate the long-term disruptive nature of technology and digital business models. Digital transformation indeed has plenty of potential for organizations, yet it does not come free. Coaching the business means telling them what they need, but often do not want, to hear. Showing them what they need to, but do not want to, see: complexity, interdependencies, interconnectedness. This is at the risk of, yet again, being seen as someone who slows down, restricts, and audits the business rather than

¹ Gartner IT Glossary, 2016, Bimodal IT: www.gartner.com/it-glossary/bimodal.

enabling it. However, true digital transformation can only happen when organizations face and deal with these issues. It is a capability they have to further develop and enterprise architects play an important coaching role in this: learning how to face, deal with, and even capitalize on these challenges. But at the risk of having to leave the digital party, enterprise architects should transcend their traditional role of operating in a constraint-mode. They have to move beyond the challenges and see the opportunities.

They lead and get out of their comfort zone.

Dealing with innovation, seeing the outside-in perspective, understanding end-clients, and understanding changing business models all seem far away from what many enterprise architects focus on today. Still, if enterprise architects restrict themselves to the domain of feasibility, they condemn themselves to being kings of constraints. Many business (transformation) managers are searching for ways to enable digital transformation and accelerate the pace. They do not need enterprise architecture roadblocks in their way. Even if these roadblocks are useful and valuable. In a digital context, there is a strong need for architectural leadership, as the organization eventually will have to be capable of scaling and industrializing the newly developed digital innovations. In order to take up this leadership, enterprise architects must evolve towards operating in an “option-mode” and “opportunity-mode”. This new role of opportunity architect does not imply that architects should opportunistically force their involvement in the transformation upon the business wherever possible. On the contrary, opportunity architects choose their battles strategically and only intervene where their involvement is of true value. As such, they provide direction in redesigning the existing organizational structures, or they create opportunities for innovation by thinking out-of-the-box. This means enterprise architects should take up a leading role in explaining, for example:

- How open APIs will help their organization lead the way in a newly emerging ecosystem
- How a data lake is crucial to better understanding customer needs
- Why poor information management increases customer churn
- Which back-office investments are needed before the business can truly scale at the pace it wants to
- Which business processes are superior to the competitor's and should be further leveraged
- How cross-organizational data-integration can build new services that can delight clients

- Which business capabilities need to be co-created with clients and can be supported by the organization's processes and IT systems

Not every enterprise architect *is fit* to become an opportunity architect. Not every enterprise architect *has* to become an opportunity architect. But what is clear from this study, is that organizations in the midst of a digital transformation need opportunity architects. Traditional enterprise architects will always be needed and will continue to have a place in organizations. But more is needed in this fast-changing business environment. Perhaps traditional enterprise architects will gradually become a commodity whilst opportunity architects will be the true source of differentiation for digital transformation.

ABOUT THE AUTHORS

Greet Bontinck is a Research Associate at Vlerick Business School, Belgium. At Vlerick, she conducts research for the Centre of Excellence in Enterprise Architecture and the Telenet Chair on Data-Driven Business Models.

Bjorn Cumps is a Professor at Vlerick Business School, Belgium, where he heads the Vlerick Centre for Financial Services and co-founded the Centre for Excellence in Enterprise Architecture. His research mainly focuses on business-ICT alignment, management of ICT, enterprise architecture, mobile technologies, new media, and how these all impact business models and operations.

Stijn Viaene is a Full Professor at Vlerick Business School, Belgium, where he heads the Information Systems Management Cluster, and co-founded the Centre for Excellence in Enterprise Architecture. He is also a Professor in the Decision Sciences and Information Management Department at KU Leuven, where he holds the VDAB Chair of Digital Business Innovation of Public Services.

Wesley Bille started his career as a researcher at the Vrije Universiteit Brussel (VUB), mainly focusing on the domain of conceptual modeling. After obtaining his PhD in 2007, Wesley left the academic world to start a career in the sector of IT Services. Currently Wesley is unit manager Digital Transformation at LoQutus.

Joachim Vanden Brande graduated in applied informatics and consequently worked as an ICT consultant in several large organizations in banking, automotive, and services. Joachim is the co-founder of LoQutus, an ICT consulting company with focus on (enterprise) architecture, and Ground Lion, a software solution supporting complex knowledge based processes.

REFERENCES

S. Bente, U. Bombosch, S. Langade: Collaborative Enterprise Architecture: Enriching EA with Lean, Agile, and Enterprise 2.0 Practices, Waltham, MA, Morgan Kaufmann (2012).

O. Bossert, C. Ip, J. Laartz: A Two-speed IT Architecture for the Digital Enterprise, McKinsey Quarterly, December 2014.

B. Cumps, S. Viaene, P. Dussart, J. Vanden Brande: Towards Enterprise Architecture Infused Organizations, Journal of Enterprise Architecture 9 (2), 8-18 (2013).

J. Gøtze: The Changing Role of the Enterprise Architect, Proceedings of the 2013 17th IEEE International Enterprise Distributed Object Computing Conference Workshops (EDOCW 2013), September 9-13, 2013, Vancouver, BC, Canada (2013).

B. Nagji, G. Tuff: Managing your Innovation Portfolio, Harvard Business Review, 5, 69-102 (2012).

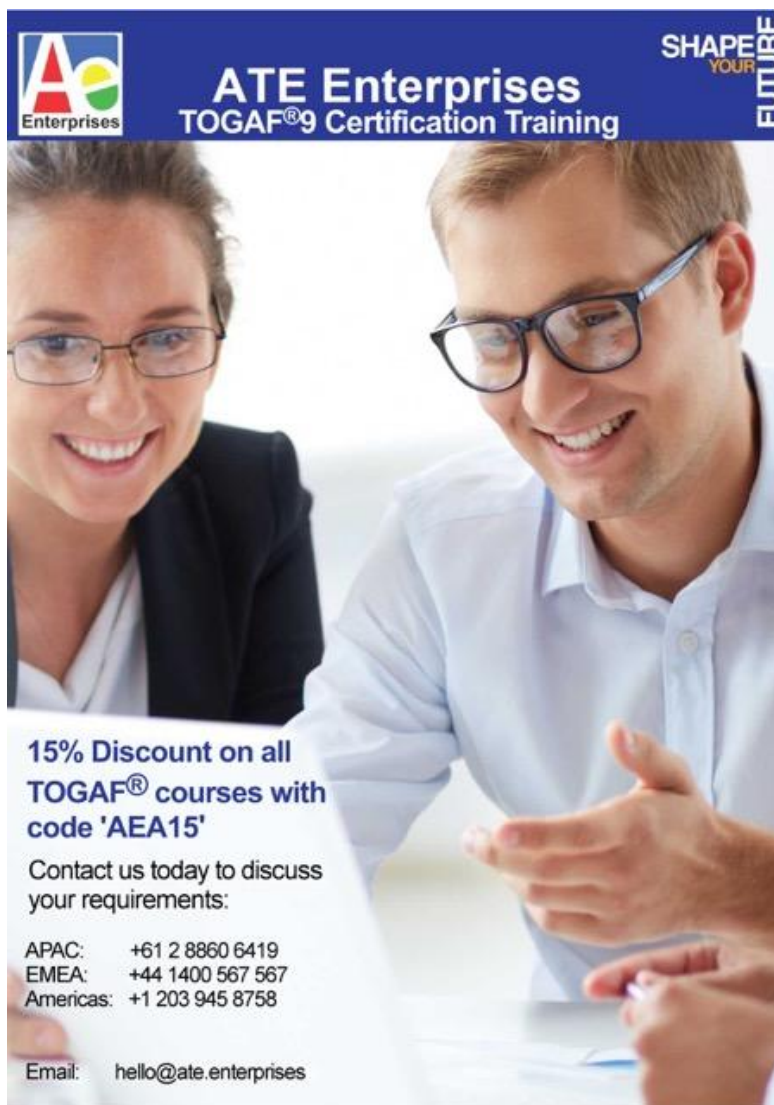
J.W. Ross, M. Mocker, I. Sebastian: Architect your Business – Not just IT!, MIT CISR Research Briefing, Vol. XIV, December 2014.

N. Safrudin, M. Rosemann, R. Jung, A. Uhl: Innovation Capability, in Digital Enterprise Transformation, A. Uhl and L. Gollenia (Eds.) Farnham, Gower Publishing Ltd. (2014).

T. Scheer, C. Burns: Enabling Supernormal Growth into Adjacent Digital Markets, Cutter IT Journal, Vol. 28, No. 11/12, 14-15 (2015).

J. Tidd, J. Bessant: Managing Innovation: Integrating Technological, Market, and Organizational Change, Chichester, England, John Wiley & Sons (2009).

A. Uhl, M. Born, A. Koschmider, T. Janasz: The Importance of Technological Trends and How to Exploit Them for Business Excellence, in Digital Enterprise Transformation, A. Uhl and L. Gollenia (Eds.) Farnham, Gower Publishing Ltd., pp.1-26 (2014).



ATE Enterprises
TOGAF®9 Certification Training

SHAPE YOUR FUTURE

15% Discount on all TOGAF® courses with code 'AEA15'

Contact us today to discuss your requirements:

APAC: +61 2 8860 6419
EMEA: +44 1400 567 567
Americas: +1 203 945 8758

Email: hello@ate.enterprises

Short Subject

Architectural Roles in an “Agile Landscape”

Phil Bowker, Jacques Colle, Kenneth van Rumste

Abstract

Agile advocates that the role of an architect is part of the process, but only very few resources can be found which describe the role and the deliverables they should create in the projects in which they are contributing. The objective of this article is to provide insights in agile and architecture, describe the role of architects, position the architect in the agile landscape, and define its deliverables.

Keywords

Architecture, architect, agile, role, deliverable

INTRODUCTION TO ARCHITECTURE AND ITS ROLES

Originating in the manufacturing and construction industries, the waterfall model was introduced into ICT projects with the objective of industrializing the process (Royce 1970).

At the outset, the science of ICT project management was both young and innovative, but a lack of general experience and know-how in the field led initially to a high percentage of project failures. To address this, and under the banner “you can’t build a roof if you don’t build quality foundations first”, in the 1990s managers of ICT projects started to apply a project management method developed at the start of the 1970s that had been proven to vastly improve the success rate of projects in industry: the waterfall model.

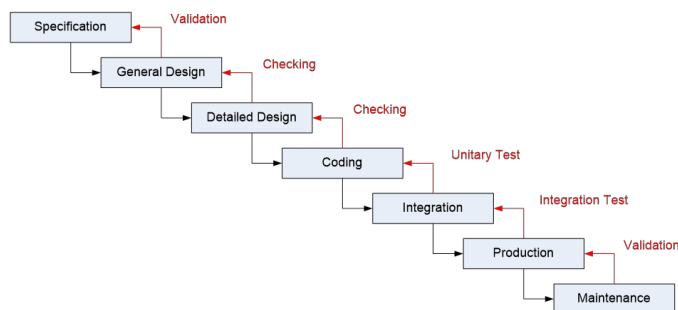


Figure 1: Waterfall Model

The waterfall model was conceived as a sequence of activities that, at the end of each step, delivers a set of documents that verifies and guarantees the quality and conformity of that completed step. This guarantee is the “stake in the ground” that allows the process to pass to the next step.

Thus, to move from a general understanding of the requirement to a more detailed understanding, the waterfall model takes us through steps that guarantee the integrity of our understanding.

However, this method has a major disadvantage, which is that any misunderstanding, deviation, or error in one step requires the process to return to the start of that step or maybe even to one of the steps taken earlier, which is costly in both time and budget.

To minimize the potential impact, ICT project managers looked to improve their processes whilst maintaining the principle that a quality foundation is the cornerstone of a successful project. As a result, new models began to appear, one of the most popular of which is the “V model” (TutorialsPoint), which allowed changes to be addressed in the corresponding opposite activity of the “V”, rather than having to work back through the entire cascade, and for validation of downstream activities to be defined upstream.

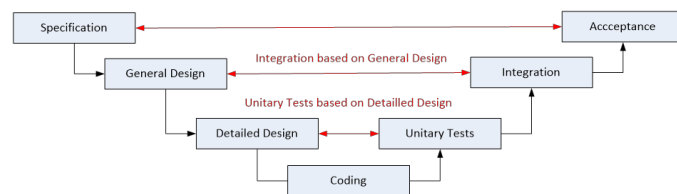


Figure 2: The “V Model”

There are several other topics of improvement concerning ICT project management, but the purpose of this article is to focus on another aspect of the search for assured quality in IT systems: Information Systems Architecture.

Though the birth of system architecture may be attributed to the work of Dewey Walker (Schekkerman 2008), the director of the department of architecture at

IBM, in the middle of the 1960s, it is the work of John A. Zachman and Steve H. Spewak with the Zachman framework and the book “Enterprise Architecture Planning”, respectively, that developed the discipline which we know today. Zachman stated the drivers for his framework were:

"The need to use a logical construction blueprint (i.e., an architecture) for defining and controlling the integration of systems and their components", and

"The bridge between the strategy and its implementation."

These drivers have the same goal as project management methodologies; i.e., to improve the quality of the development of information systems. The correlation between project management methodologies arising from the construction world and information system architecture was confirmed in 1998 by Jacques Sassoone (Sassoone 1998), who proposed to structure information system architecture by zones, districts, and blocks (analogy with the urbanization of a city (city plan)).

Today we divide the work of the architects into several roles, or functions (Longép  2001):

Enterprise architect (EA): responsible for:

- Creating and maintaining the architectural roadmap for projects
- Providing the integration strategy at enterprise and group level
- Architectural governance in the absence of specific IT governance
- Functional model within the enterprise

Domain architect: responsible for:

- Architectural integration of projects in the relevant domain (a functional entity corresponding to a business or a step of the value chain)
- Validation of the architecture model generated by the functional architect(s)
- Managing application patrimony of the domain

Functional (solution) architect: responsible for:

- Defining and producing a model for the solution
- Integration of projects within the architecture of the domain in terms of functionalities and interfaces
- Validation of interfaces with other products
- The functional architect in collaboration with the project leader is responsible for budget estimation, but the project leader is responsible for budget management

Technical architect: responsible for the technical design of the application.

Infrastructure architect: responsible for the design of the infrastructure.

The two methods (urbanization of IS system and agile method) are perfectly compatible if the enterprise and domain architects integrate the IT strategies in the different business strategies. Furthermore, the functional, technical, and infrastructure architects prepare the general design of the application (with enough information to allow the optimization of necessary costs, time, and resource estimation by the project leader) and have to control the deliverables of functional and technical analysts in charge of detailed design of the application.

Towards the end of the 1990s and into the early part of the 21st century:

- Large-scale “back-end system” development was coming to an end
- Internet and the idea of re-usable software components were well established in the enterprise
- Client-side software had taken a role at the heart of information systems, and cloud computing

These four factors significantly changed the emphasis within the enterprise with regard to the construction of information systems; flexibility became of greater value than quality. From this point, the quality of an application would be evaluated more in terms of the speed and capability to change and redeploy the application rather than its base level of quality.

To respond to these requirements project managers turned to new methods termed “agile” to show their suitability to meet the new requirements.

THE “AGILE MANIFESTO”

The Agile Manifesto (<http://agilemanifesto.org>) is a text that first appeared in 2001 written by 17 experts in software development. The text comprises four values supported by 12 principles, which are reproduced below:

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. “Agile processes” harness change for the customer’s competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference for the shorter timescale.
4. Business people and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

7. Working software is the primary measure of progress.
8. “Agile processes” promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity – the art of maximizing the amount of work not done – is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

AGILE TERMINOLOGY

(User) Story: An item on the Backlog expressing a need, including both user and technical.

Backlog: A collection of functional requirements, expressed as Epics (Portfolio Backlog), Features (Program Backlog), or User Stories (Team Backlog).

Epic: A collection of Features and/or Stories.

Feature: The implementation by the team of a service or functional requirement that provides a visible effect on the product evolution.

Release: A period of time comprising a number of Sprints. Used to provide medium-term planning and to aid release management.

Sprint: A regular, fixed period of time during which a potentially shippable increment of the product is created.

MAKING THE LINK TO AN “AGILE LANDSCAPE”

The objective of this article is not to describe in detail any method, like agile; there are a multitude of resources available in the public domain (for example, Scrum, eXtreme Programming, RAD, Chrystal Clear, ...) that satisfy this need. Rather, this article proposes a means by which “agile processes” might be employed yet maintain the advantages described previously that come from a coherent approach to IT information system architecture.

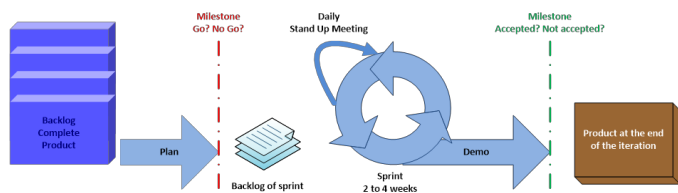


Figure 3: Agile Process

The principal technique of “agile” is to break up project activity into small pieces, permitting changing

management of priorities and a rapid feedback loop for the product increments developed (product at the end of iteration). The main problem with this approach is maintaining a coherent, holistic view of the product. For a project realized according to the “V model” this responsibility belongs to the architects.

An agile approach such as Scrum (www.scrumalliance.org) or the SAFe (Scaled Agile Framework: scaledagileframework.com) methodology, one of the most popular and widely adopted “agile processes”, focuses on optimizing incremental software deliveries, perfectly suited to website development, for example.

In this context the role of the enterprise and domain architects are not affected by the use of an “agile approach”, because such an approach changes neither the original architectural approach nor overall management of a domain or enterprise.

The principal functional (or solution) architectural deliverable is a set of models that represent a milestone in the execution of a project. This set of models verifies the functional solution for a particular project in the context of the enterprise architecture, in the following terms:

Functional: Verification of the functional coherence of the software to be developed, including identifying potential re-use.

Application: Verification of the conformance to standards promoted by the information systems governance, and of the interfaces to verify planned integration.

Technical: Verification of the use of standards and best practices in terms of technical architecture (re-usable components, protocols, etc.).

Infrastructure: Verification of the feasibility of the implementation on existing infrastructure or a new infrastructure conforming to the standards of the enterprise.

Security: Verification of the solution with respect to IT security (application, technical, infrastructure, and users).

Budgetary: Verification that the project manager has at his or her disposal all the necessary information to specify and estimate the resources, overall cost, and time required for the project.

The 11th principle of the “Agile Manifesto” states:

“The best architectures, requirements, and designs emerge from self-organizing teams”.

Taken in the Scrum context, it is the team that decides which stories shall be implemented in a Sprint, giving preference to those with highest priority as defined by

the team and taking into account the team's capacity (in Scrum terminology, its velocity). In this context the 11th principle, the Sprint, is a "system" composed of different "User Stories" (or simply "Stories").

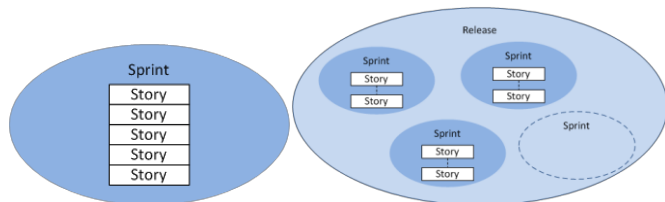


Figure 4: A "Sprint" as a System or "Release" as a System

Any optimization is therefore done at Sprint level, so the "best" solution is the best solution for that Sprint delivery and the set of stories contained therein.

In the same way, a "Release" is the set of deliverables from one or more Sprints. However, the optimization of the whole is not the same as the optimization of the constituents. The analogy with construction remains a good example; it is not because you optimize each construction activity that you optimize the building, nor that you improve the area in which you construct.

We can therefore envisage optimization at the level of the Release by the team responsible for release management, resulting in certain rules or constraints imposed on the developers and the Sprint deliverables.

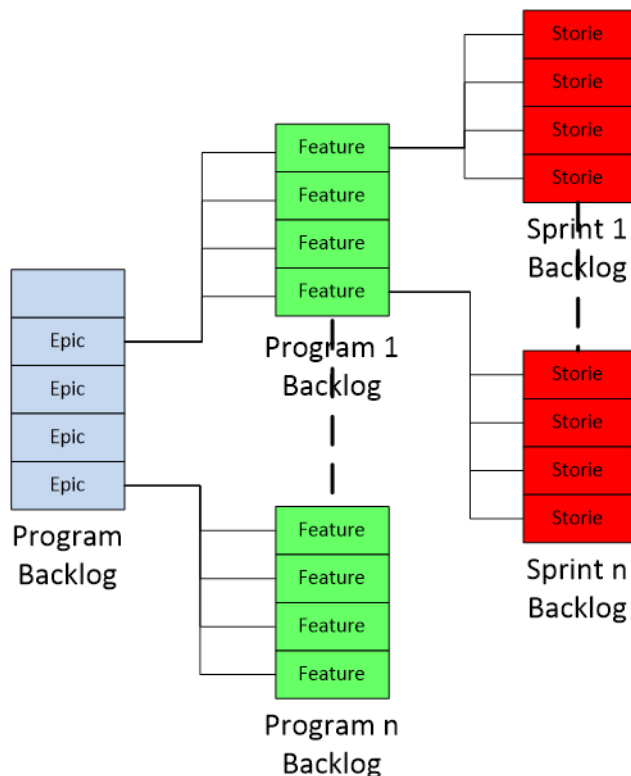


Figure 5: Backlog Hierarchy

Extending Scrum to before the start of development in a Sprint allows us to manage the steps necessary for preparation of a Sprint. For this, the product owner defines "Features" of the product, documented on a Feature Backlog. These Features are then decomposed ("sliced" in "agile" terminology) into User Stories, which are then managed on the Product Backlog before being handled on the different Sprint Backlogs.

Finally, following this new methodology, what remains is the concept of the "Epic", which is an assembly of Features and/or User Stories, and is managed at program level for a given domain.

ARCHITECTURAL ROLES IN AN "AGILE LANDSCAPE"

So as to not lose the relationship of architecture to projects in an "agile landscape", we defined the roles of the various architects in view of maintaining the necessary coherence for each product. In keeping with the construction analogy, the most important point was to guarantee a functional coherence across all aspects of the solution. With an "agile approach" it is important to conserve this coherence from the "Epics" to the "Stories". This coherence implies minimizing the occurrence of functionality in the solution, avoiding duplication, and an implementation that follows a tree structure and that ensures that a "Story" only ever applies to one "Epic".

To achieve this, we adapt the architects' roles in an "agile landscape" as follows:

Enterprise architect (EA): responsible for:

- Creating and maintaining the architectural roadmap for Epics
- Providing the integration strategy at enterprise and group level
- Architectural governance in the absence of specific IT governance
- Functional decoupling within the enterprise

Domain architect: responsible for:

- Integration of the Epics and Features in the given architectural domain
- Validation of the architecture model generated by the functional architect(s)
- Managing application patrimony of the domain

Functional (solution) architect: responsible for:

- Defining and producing a model for the solution
- Integration of projects within the architecture of the domain in terms of functionalities and interfaces (the model must guarantee a functional coherence from the Epics to the Stories, as well as coherence

with the application and infrastructure architectures)

- Validation of interfaces with other products
- Validation of “agile” deliverables with respect to standards and security rules
- Note budgetary aspects transfer from the functional architect to the product owner

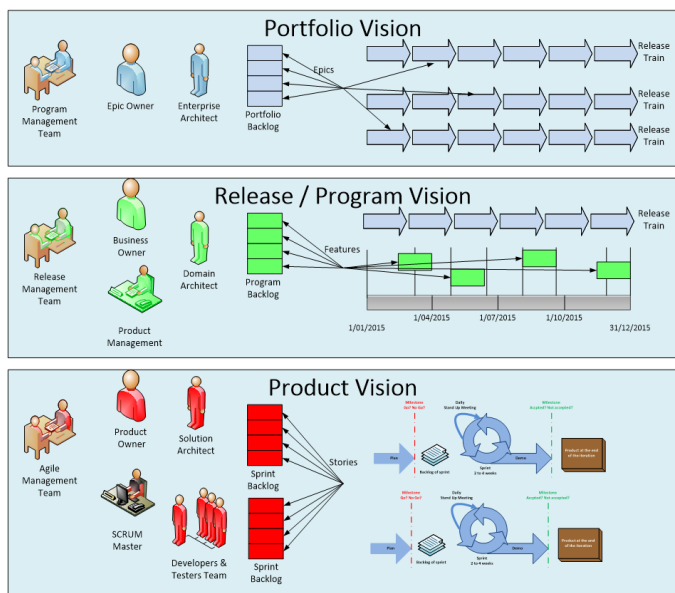


Figure 6: Global View for Architectural Roles in an “Agile Landscape”

ABOUT THE AUTHORS

Phil Bowker has over 20 years' experience as an independent IT consultant. Since his agile epiphany, Phil has spent the last 10 years helping organizations in their transformation towards agility.

Jacques Colle has over 20 years' experience as Enterprise Architect in different companies (GDF Suez, BNPPARIBASFORTIS, Elia SA, ING, etc.). Jacques is a board member of the Belgian chapter.

Kenneth van Rumste, has experience in project management and is Expert Leader Project Management.

REFERENCES

C. Longépé: Le Projet d'Urbanisation du Système d'Information, Dunod, 978-210005694 (2001).

W. Royce: Managing the Development of Large Software Systems: Concepts and Techniques, in Proceedings of IEEE WESCOM, IEEE Computer Society Press, Los Alamitos (1970).

J. Schekkerman: Enterprise Architecture Good Practices Guide, Trafford Publishing (2008).

J. Sassoon: Urbanisation des Systèmes d'Information, Hermes, 2-86601-693-9 (1998).

TutorialsPoint, SDLC – V-Model:

https://www.tutorialspoint.com/sdlc/sdlc_v_model.htm.

Book Review

Software Design Decoded: 66 Ways Experts Think, by Marian Petre & André van der Hoek, with Illustrations by Yen Quach

MIT Press, 2016, ISBN 978-0-262-03518-7

Seventy-nine Short Essays on Design, by Michael Bierut

Princeton Architectural Press, 2007, ISBN 978-1-56898-699-9

Reviewed by Leonard Fehskens

Keywords

Design, software design, graphic design, essays, experts

INTRODUCTION

I have long believed two things; first, that architecture is a specific kind of design, and thus that a prerequisite to being a good architect is to be a good designer, and second, that the more different kinds of design one is familiar with, the more likely one is to be a good designer.

These two books, one about software design, and the other about graphic design, while similar in concept are very different in execution, but I believe both may be of value to the thoughtful enterprise architect.

PARTICULARS

“Software Design Decoded” is a small book, 14.2 cm x 16.3 cm x 1.6 cm. It is unpaginated, and comprises, if I counted correctly, 94 leaves for 188 pages. There is a Preface, Acknowledgements, source notes for the illustrations, and brief biographies of each of the authors and illustrator. There is no table of contents or index and there are no references.

The 66 expositions of ways software design experts think are each presented as a two-page spread with a line drawing illustration on the left and a short paragraph on the right. They are organized into 11 groups with the following themes:

- Experts keep it simple
- Experts collaborate
- Experts borrow
- Experts break the rules
- Experts sketch
- Experts work with uncertainty
- Experts are not afraid

- Experts iterate
- Experts test
- Experts reflect
- Experts keep going

“Seventy-nine Short Essays on Design” measures 15.9 cm x 23.6 cm x 2.5 cm. It comprises 272 pages and includes a Preface, a Table of Contents, the seventy-nine essays (each from two to nine pages long) ordered chronologically by when they were first published, an Appendix with notes for each essay (including references), an Index, and a brief About the Author. This is a book about graphic design, and as such its central conceit is that each essay is set in a different typeface. The titles of the 79 essays, set in their respective typefaces, adorn the front and back covers of the book. The typefaces are identified in each essay’s entry in the Appendix.

It took me about 40 minutes to read “Software Design Decoded”. It took me about a month to read “Seventy-nine Short Essays on Design”, two or three essays at a time.

THE RELEVANCE TO ENTERPRISE ARCHITECTURE

What value can a book about software design and one about graphic design provide to an enterprise architect? Let’s assume that you share my belief that architecture is a specific kind of design, and thus that a prerequisite to being a good architect is to be a good designer. How then might these books help you become a better designer, and thus a better architect?

At times, reading “Software Design Decoded” felt like I was reading a PowerPoint presentation, with everything parceled out in bullet points. I don’t think the illustrations added much to the value of the book, and I had a hard

time imagining how some of them related to the subject at hand. The value of “Software Design Decoded” is in its succinctness, as much in what it leaves unsaid as what it does say.

Think of it as a checklist – whether you consider yourself to be a beginning designer, a competent designer, or an “expert” designer, you can ask yourself “as a designer, do I do these things, do I think these ways?”. But it is the tip of the iceberg, as it were, and it leaves it up to you to do the deeper exploration. What is “left to the reader as an exercise” is where the real value of this book lies. It’s up to you to find the answers to questions like “why is this considered a best practice for designers?”, or “how do I actually do this?” The authors and their publisher don’t leave you entirely on your own, but you have to follow an easily overlooked suggestion on the page preceding the Preface:

“For more information and background about the book, as well as additional insights contributed by the community, see mitpress.mit.edu/software-design-decoded.”

That link takes you to a conventional blurb about the book. You then have to follow a link from that page to the “companion website”, which is at softwaredesigndecoded.wordpress.com. And there, there’s a link to a fairly comprehensive bibliography (around 80 items) that really should have been in the book itself. I was disappointed to find that the site provides little “background about the book” that’s not already in the book, and more importantly there’s no evidence of the promised “additional insights contributed by the community”, though there is a form you can use to contribute such insights, wherever they might end up.

As to the “66 Ways” themselves, they’re pretty consistent with my own experience as a designer, what I’ve learned from conversations with other designers, and what I’ve learned from my explorations of the literature of design. Very few of the “Ways” are described in a software-specific manner and most are thus easily generalized to other design contexts. Something that is implicit in several of the authors’ observations that I would have elevated to distinct “Ways”, are the idea of balance and the related ideas of harmony and synergy.

As noted earlier, “Seventy-nine Short Essays on Design” takes a very different approach. Its value lies in the opportunity it provides to get a close up view of how a highly respected (graphic) designer thinks about and does the kinds of things in the “Software Design Decoded” checklist. This book requires a bit more work on the reader’s part to apply that value to the practice of enterprise architecture, but in my experience enterprise architects often have wide-ranging interests and the

graphic design context will be a feature rather than a distraction. In almost every essay there was at least one thing, and often many things, that I highlighted as a useful insight about design or the sociopolitical context within which design unavoidably takes place, or a succinct expression of a design principle.

“Seventy-nine Short Essays on Design” is not an academic book; it is a collection of very readable considerations by a thoughtful working designer. It is short on theory and long on pragmatics. To whet your appetite, some essay titles:

- Why Designers Can’t Think
- How to Become Famous
- In Search of the Perfect Client
- The Real and the Fake
- George Kennan and the Cold War between Form and Content
- Better Nation-Building through Design
- The Forgotten Design Legacy of the National Lampoon
- Information Design and the Placebo Effect
- The Idealistic Corporation
- The Tyranny of the Tagline
- What We Talk about When We Talk about Architecture
- The Supersized, Temporarily Impossible World of Bruce McCall
- Authenticity: A User’s Guide
- Designing Under the Influence
- The Mysterious Power of Context
- Wilson Pickett, Design Theorist, 1942-2006
- Design by Committee
- The Road to Hell: Now Paved with Innovation!
- In Praise of Slow Design

And there are 60 more! (BTW: Wilson Pickett, yes, Wilson Pickett the R&B singer, was born in 1941.)

Both books are relatively inexpensive, but “Seventy-nine Short Essays on Design” delivers considerably more “bang for the buck”, if you’re willing to make the effort to reap it.

ABOUT THE REVIEWER

Leonard Fehskens is the Chief Editor of the Journal of Enterprise Architecture.

Article

A Conceptual Framework for a Professional Code of Ethics for Enterprise Architects

Leonard Fehskens

Abstract

A professional code of ethics is generally considered to be an essential element of a professional ecosystem. This article summarizes the key conclusions and recommendations of the sociological literature on codes of ethics that are relevant to the development of such a code for the emerging profession of enterprise architecture, and their implications for such a code.

Keywords

Ethics, code of ethics, profession, enterprise architecture

INTRODUCTION

"Unions are, like businesses, primarily organizations of self-interest. They exist for the benefit of their members, just as businesses exist for the profit of their owners. A profession, in contrast, is organized to help its members serve others – according to a certain ideal expressed in its code of ethics." (Davis 1991)

Harold Wilensky, an organizational sociologist at the University of California Berkeley, studied what he calls the "process of professionalization" and observed the occurrence of certain common steps toward the recognition of a body of practice as a profession. These are summarized in Table 1 (Wilensky 1964).

On the path a body of practice takes to becoming a profession, the creation of a formal code of ethics represents a significant milestone.

"Eventually rules to eliminate the unqualified and unscrupulous, rules to reduce internal competition, and rules to protect clients and emphasize the service ideal will be embodied in a formal code of ethics." (Wilensky 1964)

A profession is generally characterized by expertise which is opaque to its clients, making it difficult for clients to assess a professional's mastery of the necessary skills and knowledge, and the relevance of the professional's experience. The relationship between professionals and their clients is thus inherently asymmetric, with the client both depending on and having to trust the professional.

PURPOSE OF A CODE OF ETHICS

This asymmetric relationship of trust and dependence creates opportunities for ethical dilemmas:

"It has long been recognized that the expertise and privileged position of professionals confer authority and power that could

readily be used to advance their own interests at the expense of those they serve." (Frankel 1989)

Mark Frankel (Frankel 1989) identifies eight roles that a code of ethics can play for a profession:

- Enabling document – *"as a distillation of collective experience and reflection, a code can offer guidance to individual professionals by simplifying the moral universe and by providing a framework for organizing and evaluating alternative courses of action".*
- Source of public evaluation – *"as a visible pronouncement of a profession's self-proclaimed role and values, a code can function as a basis for the public's expectations and evaluation of professional performance, thus serving as a mechanism for holding the profession and individual professionals accountable".*
- Professional socialization – *a code can "help to foster pride in the profession and strengthen professional identity and allegiance".*
- Enhance profession's reputation and public trust – *"to the extent that a code confers benefits on clients, it will help persuade the public that professionals are deserving of its confidence and respect, and of increased social and economic rewards".*
- Preserve entrenched professional biases – *"any profession has a clear, tangible stake in protecting its members' professional monopoly, and a code of professional ethics can help secure such protection".*
- Deterrent to unethical behavior – *"in addition to promoting ethical conduct, a code may serve as a deterrent to unethical behavior ...".*

- Support system – “a code may also constitute a legitimate source of support for professionals against unwarranted erosion of their power or improper demands on their skills by outsiders”.
- Adjudication – “codes may serve as a basis for adjudicating disputes among members of the profession or between members and outsiders”.

The literature on professional codes of ethics tends to emphasize the “enabling document”, “enhance profession’s reputation and public trust”, and “deterrent to unethical behavior” roles.

This literature shows a surprising variety of ways the community of nominal experts on professional codes of ethics thinks about the subject. Some commentators draw distinctions between codes of ethics, standards of conduct, and codes of practice (see, for example, Davis (2009), p.52). Others observe that, in practice, things called a “code of ethics” often mix these up, sometimes with other things as well, especially material about individual character reminiscent of the Boy Scout Law (BSA):

A Scout is: Trustworthy, Loyal, Helpful, Friendly, Courteous, Kind, Obedient, Cheerful, Thrifty, Brave, Clean, and Reverent.

John Kultgen, Professor of Philosophy at the University of Missouri, has compiled a table of the subjects addressed by several representative professional codes of ethics (Kultgen 1988). This table is reproduced in Appendix A.

In general, the less well bounded a concept of a code of ethics is, the more likely the code of ethics will be lengthy, complex, and lacking coherence. This will in turn increase the likelihood that the code of ethics can be interpreted in different ways, making it difficult to use and enforce.

“A code that means different things to different people can do more harm than good.” (Kultgen 1988)

These observations suggest that a code of ethics for the enterprise architecture profession should focus on the “enabling document role”, dealing with the specific kinds of ethical challenges a practicing enterprise architect might encounter, and addressing the public image and deterrent roles as secondary consequences of this focus.

The literature of professional ethics is clear on one thing – a code of ethics must be unambiguous, understandable, and relevant.

WHAT MAKES AN ACTION ETHICAL OR UNETHICAL?

Scholarly discussions of ethics typically draw a distinction between the ideas of legal, moral, and ethical behavior. All three are rooted in the notion that

behaviors that lead to outcomes that are inconsistent with or incompatible with the prevailing values of a community (ICAF) are to be discouraged. Different scholars have different ideas about how these ideas are best differentiated, and the philosophical arcana of such debates are far beyond the scope of this discussion. The distinctions drawn by Geoffrey C. Hazard (Emeritus Professor of Law and Adjunct Professor of Law at the University of Pennsylvania) are as good as any I have encountered, and have the advantage of being clearly and succinctly expressed (Hazard 1995).

“By ‘law’ I mean norms formally promulgated by a political authority that are enforceable and more or less regularly enforced through a legal process based on adjudication.

By ‘morals’ I refer to the notions of right and wrong that guide each of us individually and subjectively in our daily existence.

By ‘ethics’ I mean norms shared by a group on a basis of mutual and usually reciprocal recognition. Ethics, as thus defined, is essentially a two-party transaction.”

Hazard uses these definitions to make the following distinctions:

“The point is to recognize that moral notions are single-person events that animate each individual’s life and cannot be fully communicated to others. In their very nature, therefore, subjective moral standards cannot be publicly expounded like legal rules, nor can they be objectively administered as legal rules can be.

Shared mutual intelligibility distinguishes what I am calling ‘ethics’ from what I have called ‘morals’.”

The kinds of prevailing values of a community that are relevant to ethical behavior (ICAF) – i.e., how we should treat one another – typically embrace such things as honesty, fairness, justice, trustworthiness, loyalty, civility, nonmaleficence, etc.. Many of these values reflect two “common sense” ideas about ethical behavior: reciprocity, as implicit in the “Golden Rule” – “do unto others as you would have them do unto you”, and “first, do no harm”, often attributed to the Hippocratic Oath.

WHAT SHOULD THE SCOPE OF A PROFESSIONAL CODE OF ETHICS BE?

A professional code of ethics cannot be a universal moral code. It is, almost by definition, limited to the kinds of behaviors that are expected of members of a particular profession, as members of that profession, in ethical contexts that are specific to that profession. So, for example, it is probably not necessary for a professional code of ethics to explicitly require that the members of a particular profession not break the law.

For a comprehensive survey of the kinds of ethical considerations relevant to professionals, see Johnson (2008).

STAKEHOLDER CLASSES

While the primary motivation for a code of ethics is the asymmetric relationship between the professional and the client, others participating in or affected by the practice of enterprise architecture may be involved in or affected by an ethical dilemma. These include:

- Individual Practitioners – the members of the profession of enterprise architecture, to whom a professional code of ethics would apply.
- Practitioners' Colleagues – people with whom practitioners work and, in particular, collaborate.
- Practitioners as a Profession – the profession of enterprise architecture as a whole.
- Employers of Practitioners – the people who directly compensate (i.e., provide pay and benefits to) practitioners. A practitioner may be self-employed.
- Clients of Practitioners – the people or organizations on whose behalf practitioners work; the consumers of practitioners' work products.
- Society – Society at large, especially the great majority who are largely unaware of or indifferent to the practice of enterprise architecture.

BAD THINGS

The occurrence of an ethical infraction may have different kinds of outcomes that are generally considered unfair or unethical:

- Personal advantage, accruing to the individual practitioner considered to be responsible for the ethical infraction
- Client disadvantage or harm
- Third-party disadvantage or harm (this third party may be another enterprise architecture practitioner, a member of a different profession, a non-professional, an organization, the enterprise architecture profession, another profession, society, ...)
- Third-party advantage

EXAMPLES

Combining these kinds of outcomes with these kinds of mechanisms, one can generate some examples of the kinds of ethical infractions a code of ethics might identify as unethical:

- Using privileged ("insider") information about a client's strategy or imminent actions to make highly profitable stock market trades
- Selling innovative product or process information to a client's competitor

- Observing unethical activities on the part of colleagues and failing to report it to the appropriate people
- Using the capital resources of a client for a personal project or business venture
- Providing advice or recommendations to someone that are unfairly advantageous to a third party in exchange for their preferential treatment in other business dealings
- Appropriating someone else's ideas or work and presenting it as your own

CONSEQUENCES

Codes of ethics which cannot be or are not enforced in some way are nothing more than rhetorical gestures. While a professional body generally cannot prevent its members from violating its code of ethics, it can provide sanctions to be applied when the code has been violated. Such sanctions are generally graduated to be proportional to the severity of the ethical violation. They include:

- Admonishment or censure – the practitioner receives a formal notification of the professional body's disapproval of the practitioner's ethical violation. This notification may also be published so that it is visible to the professional body's membership and possibly the general public.
- Suspension of membership – the practitioner's rights and privileges as a member of the professional body are suspended for some period of time. This will typically include censure.
- Suspension of professional credentials – the professional body recommends to some other accrediting or certifying body that the practitioner's professional credential be suspended for some period of time. This will likely require that an agreement between the two organizations has been previously negotiated. This recommendation may be in addition to suspension of the practitioner's professional association membership.
- Revocation of membership – the practitioner's rights and privileges of membership are revoked and future membership in the professional body is disallowed.
- Revocation of professional credentials – the professional body recommends to some other accrediting or certifying body that the practitioner's professional credential be revoked. This will likely require that an agreement between the two organizations has been previously negotiated. This recommendation may be in addition to revocation of the practitioner's professional association membership.

GOVERNANCE

Implementing a code of ethics requires several operational capabilities:

- Commitment to the code – it must be possible for members of the professional body to commit to the code in some substantive way.
- Detecting and reporting violations – while the detection of ethical violations is unavoidably dependent on the vigilance of individuals, there must be a mechanism for them to report suspected ethical violations. There is general agreement that both the reporting and subsequent assessment of suspected ethical violations should not have adverse consequences for those involved. The deliberate false reporting of ethical issues should be treated as an ethical issue itself.
- Investigating and assessing alleged violations – there must be an objectively defined process for investigating suspected ethical infractions and drawing conclusions about their nature and severity.
- Appeals – no process is infallible. There must be means for a member found to have violated the code of ethics to request an appeal and review of the decision.
- Rehabilitation and reinstatement – in some cases, it may make sense to offer a member found to have violated the code of ethics a means to make restitution and demonstrate having “learned the lesson” via a process of rehabilitation and reinstatement.
- Record keeping – all of the above require that accurate and readily accessible records be kept, with appropriate confidentiality controls.
- Amending the code – it is unlikely that the first version of a code of ethics will “get it all right”. It must be possible for the membership to propose, debate, and agree or decline to adopt changes to the code.

MOVING FORWARD

“What is the value of codification?”

The most obvious function is guidance for those practitioners who have not thought through moral issues. Codification would be otiose if the obligations of professionals were always obvious and could be met without sacrifice, if professionals never faced difficult dilemmas or always did what was right as a matter of course, or could handle every question on the basis of common morality. Debates over confidentiality, conflict of interest, bribery, paternalism, whistle-blowing, credit for the work of others, social responsibilities, etc. show that these conditions are not met. Hard thinking and searching dialog are necessary. A rational code would contain the results individuals would have reached themselves if they had

reasoned objectively long enough on an adequate base of experience. If such a code is available, it relieves professionals of most of the burden of ethical inquiry. After all, their primary responsibility is to heal, counsel, minister, design, etc., not to puzzle over ethical questions. Most in matter of fact do not reflect on ethics until faced with difficult questions, when the pressure of action prevents careful thought. It would simplify their moral universe to have solutions at hand which had been worked out on the collective experience and wisdom of the moral community.” (Kultgen 1988)

Drafting a code of ethics for an emerging profession is an important design problem fully deserving the application of all the skills and discipline we can bring to bear on the problem as enterprise architects. It is not something we can or should take lightly. As noted earlier, a code of ethics is a foundational milestone in the development of a profession that creates the cultural context for both how the profession sees itself and how it is seen by others.

A code of ethics must be tailored to the specific situations that practitioners adopting it can expect to find themselves in.

“If the basic conception of the profession’s values, roles, and functions is inconsistent with the reality experienced by its members, the basis for the respect of the profession’s ethics will erode ...” (Frankel 1989)

This salient observation suggests that the first step to developing a code of ethics for the enterprise architecture profession is to map the profession’s “opportunity space” for ethical violations.

Enterprise architecture is a profession whose clients are typically organizations rather than individuals. This will also affect the kinds of ethical situations enterprise architects may encounter.

While analogies with building architecture and city planning are commonly used as a basis for talking about enterprise architecture, the medium of expression of enterprise architecture is fundamentally different from those of other disciplines that are superficially similar. Thus, the understandable temptation to develop a code of ethics for the enterprise architecture profession by revising a code of ethics for a seemingly similar or related profession is to be resisted, especially as such use may constitute plagiarism or theft of intellectual property.

“The process used to develop a code is as important as the code itself. A review of similar codes that have already been developed can be useful to provide a starting point. However, simply copying an existing code will not address the specific needs of an organization or a professional body, neither will it be a convincing document for that organization’s membership.” (Berleur 2004)

The analysis in earlier sections of this article suggests that a feasible way to organize and populate the map of

the enterprise architecture profession's "ethical opportunity space" is to consider it as having the dimensions of:

- Who: the stakeholder(s) whom the ethical dilemma affects or involves
- Why: the "motivation" for the ethical dilemma (personal advantage, client disadvantage, third-party advantage, third-party disadvantage)
- What: the enterprise architecture medium by which the advantage or disadvantage is conferred (e.g., information, money, capital resource, decision-making influence or authority, ...)
- How: the mechanism by which the ethical dilemma arises (commission, omission, incompetence, unforeseeable/unintended side effect)

Finally, there is general agreement that the development of a code of ethics should be as participatory a process as possible.

"The widespread participation of members in such an effort helps to reinvigorate and bring into sharp focus the

underlying values and moral commitments of their profession." (Frankel 1989)

"A code that has been democratically produced is more likely to be accepted and honored by those subscribing to it than one that is delivered as a 'fait accompli'." (Berleur 2004)

"As long as the process remains insulated from public evaluation, it is at least questionable, if not unlikely, that it can produce a code that will be optimally understanding of and sensitive to the perspective of those whom the profession is pledged to serve." (Frankel 1989)

A professional body can certainly provide the means by which its members can participate in and contribute to such an undertaking, especially agreeing to adopt the code as a professional body, and to agree to conform to it individually. The participation of other stakeholders is more problematic. Thus, a fundamental "design problem" for the development and adoption of a code of ethics is to decide to what extent the participation of these other stakeholders is seen as necessary or desirable, and how this participation might be achieved.

TABLE 1
THE PROCESS OF PROFESSIONALIZATION*

	Became Full-Time Occupation	First Training School	First University School	First Local Professional Association	First National Professional Association	First State License Law	Formal Code of Ethics	No. of Errors	No. of Ties	Per Cent Error, by Groups†
Established:										
Accounting (CPA).....	19th cent.	1881‡	1881‡	1882	1887	1896	1917	0	1	
Architecture.....	18th cent.	1865	1868	1815	1857	1897	1909	2	0	
Civil engineering.....	18th cent.	1819	1847	1848	1852	1908	ca. 1910	0	0	
Dentistry.....	18th cent.	1840§	1867	1844	1840§	1868	1866	3	1	
Law.....	17th cent.	1784	1817	1802	1878	1732	1908	2	0	
Medicine.....	ca. 1700	1765	1779	1735	1847	Before 1780	1912	2	0	21
Others in process, some marginal:										
Librarianship.....	1732	1887	1897	1885	1876	Before 1917	1938	2	0	
Nursing.....	17th cent.	1861	1909	1885	1896	1903	1950	1	0	
Optometry.....		1892	1910	1896	1897	1901	ca. 1935	1	0	
Pharmacy.....	1646	1821§	1868	1821§	1852	1874	ca. 1850	2	1	
School teaching.....	17th cent.	1823	1879	1794	1857	1781	1929	3	0	
Social work.....	1898(?)	1898	1904	1918	1874	1940	1948	1	0	
Veterinary medicine.....	1803	1852	1879	1854	1863	1886	1866	3	0	27
New:										
City management.....	1912	1921	1948	After 1914	1914	None	1924	2	0	
City planning.....	19th cent.	1909‡	1909‡	1947	1917	1963	1948	2	1	
Hospital administration ..	19th cent.	1926‡	1926‡		1933	1957	1939	2	1	29
Doubtful:										
Advertising.....	1841	1900(?)#	1909(?)#	1894	1917	None	1924	1	0	
Funeral direction.....	19th cent.	ca. 1870	1914	1864	1882	1894	1884	3	0	29
							Total errors..	32		25

* Dates concern only events in the United States. Among the sources: *Occupational Licensing in the States* (Chicago: The Council of State Governments, 1952); *Encyclopedia of Associations* (3d ed.; Detroit: Gale Research Co., 1961); L. E. Blauch (ed.), *Education for the Professions* (Washington, D.C.: Government Printing Office, 1955); *Encyclopedia of the Social Sciences* and other encyclopedias; J. W. Kane, *Famous First Facts* (New York: H. W. Wilson Co., 1950); professional association journals, newsletters, and yearbooks; and specialized histories, official or not. In cases of disagreement, precedence was given to a competent history—e.g., R. W. Habenstein and W. M. Lamers, *The History of American Funeral Direction* (Milwaukee, Wis.: Bulfinch Printers, 1955)—or to a date supplied by a professional association cross-checked by one other independent source. I am grateful to Anne Mooney, Ted Cooper, and the headquarters of the dominant professional associations for assistance.

† The total number of dates out of order in group divided by total possible entries in group. The errors for the whole table are 25 per cent of possible entries.

‡ Dates in italics in the same row designate the same event.

§ Two dates in the same row marked with a section mark (§) designate associated events.

|| Only three or four physicians are known to have resided in the Colonies prior to 1700. From 1607 to 1730 Colonial medical practice was relatively primitive (R. H. Shyrock, *Medicine and Society in America: 1660-1860* [Ithaca, N.Y.: Cornell University Press, 1962], pp. 7, 18).

"(?)" designates best inference from available information.

Table 1: The Process of Professionalization (Wilensky 1964)

APPENDIX A: CONTENT ANALYSIS OF SAMPLE PROFESSIONAL CODES OF ETHICS (KULTGEN 1988)

	AAUP	ABA	AICPA	AMA	APA	ECPD	IEEE	SDC
1.	X	X	X	X	X	X	X	X
2.						X	X	
3.	X				X			X
4.		X						X
5.	X	X	X	X	X	X	X	X
6.	X							
7.	X			X	X			
8.					X		X	
9.	X	X	X	X		X	X	
10.		X		X	X	X		X
11.		X	X			X	X	
12.					X	X	X	X
13.					X	X		X
14.	X	X						
15.	X			X		X		

Table A-1: Content Analysis of the Codes

1.	Competence, objectivity, honesty about qualifications.
2.	Continuing professional development.
3.	Enlargement of knowledge, research, publication.
4.	Making services available to those who need them.
5.	Loyalty to clients, employer, or institution, avoidance of conflict of interest, bribery.
6.	Respect for autonomy, privacy, confidentiality, trade secrets.
7.	Non-exploitation of clients, students, research subjects, and animals.
8.	Proper treatment of subordinate, fairness, encouraging professional development, giving credit.
9.	Territorial rights of colleagues, cooperation, giving proper credit, fair competition.
10.	Monitoring colleagues, reporting malfeasance, preventing unauthorized practice.
11.	Protecting honor and dignity of profession, publicizing its merits.
12.	Avoidance of harm to outsiders, the public, the environment.
13.	Truthfulness in public statements.
14.	Effort to reform institutions that bear on profession.
15.	Involvement in community activities not related to profession.

Table A-2: Row Headings

AAUP	American Association of University Professors: Statement on Professional Ethics
ABA	American Bar Association: Model Code of Professional Responsibility
AICPA	American Institute of Certified Public Accountants: Rules of Conduct
AMA	American Medical Association: Principles of Medical Ethics; and Current Opinions of the Judicial Council
APA	American Psychological Association: Ethical Principles of Psychologists
ECPD	Engineers Council for Profession Development: Code of Ethics for Engineers, with Guidelines
IEEE	Institute of Electrical and Electronics Engineers: Code of Ethics
SDC	Sigma Delta Chi [Society of Professional Journalists]: Code of Ethics

Table A-3: Column Headings

ABOUT THE AUTHOR

Leonard Fehskens is the Chief Editor of the Journal of Enterprise Architecture.

REFERENCES

Jacques Berleur et al.: Criteria and Procedures for Developing Codes of Ethics or of Conduct, IFIP-SIG9.2.2 (September 2004):

www.info.fundp.ac.be/~jbl/IFIP/Criteria_and_procedures.pdf.

BSA: <http://usscouts.org/advance/boyscout/bssoathlaw.asp>.

Michael Davis: Thinking Like an Engineer: The Place of a Code of Ethics in the Practice of a Profession, Philosophy and Public Affairs, Vol. 20, No. 2 (Spring 1991):

www.erie.buffalo.edu/REU/REU%20readings/Davis-ThinkingLikeEngineer-PPA.pdf.

Michael Davis: Code Making: How Software Engineering Became a Profession, Center for the Study of Ethics in the Professions, Illinois Institute of Technology (2009):

<http://ethics.iit.edu/sea/1/book/Full%20book.pdf>.

Mark S. Frankel: Professional Codes: Why, How, and with What Impact?, Journal of Business Ethics, Vol. 8: 109-115 (Feb/Mar 1989).

Geoffrey C. Hazard Jr.: Law, Morals, and Ethics, Faculty Scholarship Series, Paper 2372 (1995):

http://digitalcommons.law.yale.edu/fss_papers/2372.

Strategic Leadership and Decision-Making, Industrial College of the Armed Forces (ICAF), Chapter 15: Values and Ethics.

W. Brad Johnson, Charles R. Ridley: The Elements of Ethics for Professionals, Palgrave Macmillan (2008).

John Kultgen: Ethics and Professionalism, University of Pennsylvania Press (1988); Chapter 10: Professional Codes.

John T. Sanders: Honor Among Thieves: Some Reflections on Professional Codes of Ethics, from Fritz Allhoff & Anand J. Vaidya, Professions in Ethical Focus: An Anthology, Broadview Press (2008):

<http://people.rit.edu/jtsgsh/PAPERS/Honor%20Among%20Thieves%202.pdf>.

James G. Skakoon (original by W. J. King): The Unwritten Laws of Engineering, ASME Press (2001); Appendix: Representative Professional Codes of Ethics.

Harold L. Wilensky: The Professionalization of Everyone?, Institute of Industrial Relations, University of California (1964). Reprinted from The American Journal of Sociology, Vol. LXX No. 2 (September 1964).

